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Costing parenting and cash transfer interventions in an early childhood development pilot programme in Tanzania

Kizazi Kijacho Phase 1 costing study

Report (updated)



thrive

This is a costing study of the Kizazi Kijacho parenting and unconditional cash transfer intervention in Dodoma, Tanzania — a pilot programme that aims to improve early childhood development outcomes in children under 2 years old.

Note that this report replaces an earlier version, first published in July 2025. It has been updated to present Phase 1 costs using a more standard distinction between investment and recurrent costs, including annualising pre-intervention costs (such as for design and initial training expenditures). The earlier report had differentiated costs by the programme implementation phase for simplicity (pre-implementation design phase and implementation phase). The update also reflects revised programme dosage and supervision data shared by the research team from the Kizazi Kijacho app.

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Acronyms and abbreviations

CC	county council
CCHWCO	council community health worker coordinator
CG	caregiver
CHW	community health worker
DC	district council
DSA	daily subsistence allowance
ECD	early childhood development
EGPAF	Elizabeth Glaser Pediatric AIDS Foundation
pcpm	per child per month
ECD	early childhood development
HCW	healthcare worker
HD	health dispensary
KK	Kizazi Kijacho
PSSN II	Productive Social Safety Net Project II
RCHWCO	regional community health worker coordinator
RCT	randomised control trial
TASAF	Tanzania Social Action Fund
TGHS	Tanzania government health scale
TZS	Tanzanian shillings
UCT	unconditional cash transfer
USD	US dollars

Executive summary

This report is a costing study of the first phase of the Kizazi Kijacho (KK, 'the next generation') randomised control trial (RCT) of a parenting and unconditional cash transfer (UCT) intervention in the Dodoma region of Tanzania.¹ This is a pilot programme that aims to improve early childhood development (ECD) outcomes among children up to 2 years old. The first phase of the pilot (the scope of this costing study) ran from March 2021 to February 2024, when the children in the study were 1 year old on average.

The KK RCT is being implemented in all districts in Dodoma and is targeted at 3,588 families (pregnant women and their newborn children up to the age of 2). The parenting intervention is delivered by existing community health workers (CHWs) via an enhanced home visit programme and new group parenting sessions. Both CHWs and their supervisors (healthcare workers, HCWs) have been trained to use a digital app to support the delivery of the parenting service. The UCT payments are delivered via mobile money to the target families. The UCT recipients are assigned to receive 3 different amounts – a high amount (USD 23.20 per month), a medium amount (USD 16.44 per month) and a low amount (USD 7.03 per month).

The costing of the KK pilot is important for three main reasons:

- costs will be needed for economic evaluation of the pilot programme in due course, allowing for comparison of cost effectiveness of the different interventions
- it uncovers the key cost drivers in the pilot programme and thus helps others thinking of designing similar pilots
- although the costs per target child in the pilot are not generalisable to a scaled-up programme, they provide a starting point for decisions about design changes or delivery adaptations that would be needed if the government were to scale the programme.

¹ A further report will investigate the costs of replicating and scaling the KK programme through government systems.

What does the study do?

For each of the three KK treatment arms – parenting only, UCT only (three different payment amounts), and combined parenting/UCT – the report answers the following research questions:

- What is the cost of the resources needed to deliver the intervention under the NGO-supported delivery model? And what is the recurrent cost per child, after the investment costs are annualised²?
- What are the main cost components and their drivers?
- How is the cost burden distributed among sources?
- What is the incremental cost of the intervention compared to the counterfactual (i.e. the cost of relevant services being delivered in the control group areas)?

To increase the usefulness of the study, the analysis extends beyond the pilot results (i.e. the cost per target child in the KK pilot programme) to consider some implementation scenarios where the same NGO-supported KK delivery model serves additional families.

The study takes a societal perspective on costs. This means that it attempts to capture all the main resources required to make the interventions work (i.e. the primary costs) regardless of who pays for them or if they are provided in-kind. This includes estimating the opportunity cost of the target children's caregivers' time in participating in the KK interventions.

The study's key data sources are financial statements from the NGOs that are implementing the KK pilot programme, as well as a small field survey of HCWs, CHWs, and caregivers undertaken specifically for this study.

What did the study discover?

- Under the NGO-supported delivery model, the combined parenting/UCT arm had the highest overall programme recurrent cost, estimated at USD 56.72 per target child per month (pcpm), while the parenting only arm cost USD 39.19 pcpm. The three UCT arms cost USD 22.72 pcpm (high payment), USD 17.90 pcpm (medium) and USD 10.38 pcpm (low). These figures reflect the intensive nature of the parenting support, which includes training, supervision, digital infrastructure and stipends for CHWs and HCWs. The combined arm's higher cost reflects the addition of UCTs, while the UCT-only arm's lower costs were largely due to the limited direct programme inputs beyond the cash transfer itself.

² Annualisation is the process of converting large, one-time capital expenditures into an equivalent uniform annual amount throughout the duration of its lifespan. It is a useful way of comparing investments that have differing lifespans. It includes expenditure on things like IT equipment, software development and one-off training.

- Training emerged as the dominant cost component in the parenting-related arms, accounting for between 24 to 31% of provider costs, depending on the parenting arm, driven by high-frequency residential sessions and daily subsistence allowance (DSA) rates. Other significant contributors included indirect programme management and programme design, which captured the costs of technical staff and support functions across the implementing NGOs.
- In the UCT-only arms, transfers were the largest component of provider costs, highlighting the simplicity and low administrative burden of that approach. Indirect management costs, however, remained notable even in the UCT-only arms, primarily due to administrative support and fixed overheads charges from the NGOs.
- Most of the costs of implementing the KK interventions were borne by the two implementing NGOs – Elizabeth Glaser Pediatric AIDS Foundation (EGPAF) and D-tree International (D-tree). They accounted for more than 95% of total costs in the parenting-related arms and nearly 100% of total costs in the UCT-related arms. The government’s financial contribution, primarily in the form of staff time from HCWs and local officials, was relatively modest in the parenting and combined arms (3–4% of total costs). These findings are unsurprising since the KK programme is an NGO-supported model that leverages existing government health functionaries for delivering parenting interventions while the UCT component is fully managed by NGOs.
- The incremental costing analysis compared the cost of implementing the KK interventions in the treatment areas against the existing services provided in the control areas. It estimated the additional resources required to deliver the programme beyond what is already offered through routine government services. Findings show that the incremental cost per target child per month was nearly equal to the KK intervention’s costs for the parenting arms, reflecting the fact that intensity of similar parenting interventions in control areas was minimal – less than 10% of treatment levels for home visits and no group parenting sessions. Incremental costs for the UCT are identical to the KK UCT costs because the government’s cash transfer programme operates in both treatment and control areas, and hence the KK UCT intervention is additional in treatment areas.
- The study revealed that the stipends and allowances paid to CHWs and HCWs generally exceeded the estimated direct labour and transport costs they incurred in delivering the intervention at the dosage reported. While this may suggest scope for increasing intervention dosage to some extent without additional cost, these decisions need to carefully consider multiple other factors such as CHW motivation and equity, particularly in extremely remote and dispersed rural community contexts.
- A sensitivity analysis of some of the key cost components was done to test the implications of key assumptions made. The costs were moderately sensitive to assumptions around NGO staff time allocation and indirect cost apportionment between the different study arms, and to the level of training allowances. Overall though, the cost estimates seem reasonably stable under the current assumptions.

- The study also ran some scenarios to look at the scope for serving more families (scaling) by improving cost efficiency under the NGO-supported model. For the parenting arm, it assessed how many families CHWs could reach under current dosage levels if they were to deliver the programme within their paid stipend or the number of hours they reported being willing to work. The findings suggest that CHWs could reach around twice as many families without exceeding either their stipend or their preferred time commitment, indicating potential for improved efficiency in a scaled model. Of course, any adaptation needs to carefully consider local contexts to maintain service quality and worker motivation.

Table 1: Summary of findings

Research question 1: What is the cost of the resources needed to deliver the intervention under the NGO-supported delivery model (constant USD 2023)?					
	Parenting only	Parenting and UCT	UCT only – high	UCT only – medium	UCT only – low
Total programme cost (total cost per target child)	832.39	1079.26	319.60	252.08	146.87
o/w investment costs	278.62	280.12	1.50	1.50	1.50
o/w recurrent costs	541.90	786.90	317.52	250.00	144.79
o/w donated	11.87	12.23	0.58	0.58	0.58
Total recurrent provider cost (cost per target child per month)	38.71	56.21	22.68	17.86	10.34
Total recurrent programme cost (cost per target child per month – includes donated costs)	39.19	56.72	22.72	17.90	10.38

Research question 2: What are the main cost drivers?					
	Parenting only	Parenting and UCT	UCT only – high	UCT only – medium	UCT only – low
Top 3 cost components	Training, 31% Indirect mgmt, 23% Direct Prog mgmt, 21%	Training, 24% Indirect mgmt, 22% Direct prog mgmt, 18%	Transfers, 77% Indirect mgmt, 15% Direct prog mgmt, 7%	Transfers, 71% Indirect mgmt, 19% Direct prog mgmt, 9%	Transfers, 51% Indirect mgmt, 33% Direct prog mgmt, 15%

Research question 3: How is the cost burden distributed among sources?					
	Parenting only	Parenting and UCT	UCT only – high	UCT only – medium	UCT only – low
NGO	97%	96%	100%	100%	100%
Government	3%	4%	-	-	-
Donated (by CHW/CG)	<1%	<1%	<1%	<1%	<1%

Research question 4: What is the incremental cost of the intervention compared to the counterfactual (i.e. the cost of relevant services being delivered in the control group areas)? Incremental recurrent cost per child per month (constant 2023 USD)					
	Parenting only	Parenting and UCT	UCT only – high	UCT only – medium	UCT only – low
Lower bound	37.17	54.70	22.72	17.90	10.38
Upper bound	37.33	54.86			

- Partly to serve as a link to the second KK costing report, which will look at the total cost of the KK programme, the final section includes estimates of overall ECD-sensitive public budgets in Tanzania, using findings from a 2025 Thrive study.³ The government currently allocates about USD 5 per month per child for all ECD-sensitive services.⁴ As would be expected for a pilot intervention operating at a small-scale supported by NGOs, the recurrent cost per child per month of the KK pilot interventions are much higher. The lowest cost KK intervention is the low-amount UCT arm at about USD 10 per child per month. A full-scale government-led intervention to a large population is likely to be leaner and benefit from existing systems and economies of scale, which will bring the cost per child down – but adaptations will be needed to make a KK model affordable for government.

What do the findings mean?

The costing study provides several important insights for the sustainability, scalability, and cost-efficiency of the KK programme.

- Among the three treatment arms, the UCT-only intervention emerged as the least costly, while the combined parenting/UCT intervention was the most expensive. On a per-child basis, even the highest payment UCT arm cost less than half that of the combined intervention, and these findings remain consistent when looking at incremental costs also.

³ Nayanpally, S., Maliti, E., Bain, N. and Rawle, G. (2025). *Public financing of ECD in Tanzania: Situation analysis to inform scaling of ECD initiatives*. Thrive, Oxford Policy Management. https://thrivechildevidence.org/wp-content/uploads/2025/04/Public-financing-of-ECD-in-Tanzania_Situational-analysis_Final.pdf

⁴ This is an indicative estimate, as detailed activity level budget data was not available and so the study team had to make assumptions on apportioning relevant budgets.

This suggests that, under the NGO-delivered model, for the combined arm to be considered a worthwhile investment, it must deliver substantially better ECD outcomes than the UCT alone.

- To transition to a government-led model at scale, the current cost per child of the pilot KK programme must be brought down significantly.

As with many NGO-led pilot programmes, KK's unit costs are much higher than existing government budget allocations for ECD services. Achieving financial sustainability will therefore require both scale efficiencies and adaptations to the delivery model.

- The main cost components of the KK programme and its drivers can help inform the design of a more cost-efficient government implementation model.

For the parenting intervention, training and indirect NGO management costs accounted for the largest shares. Alternative training modalities could reduce costs while maintaining training effectiveness (such as local, non-residential sessions, blended learning, or digital tools). Reducing indirect management costs is less problematic as these are likely to be lower under government systems due to existing infrastructure and scale. Other features of the NGO model, such as the digital app, may be harder to roll out in the public sector without considerable adaption and integration to existing digital systems.

- While stipends and travel allowances paid to CHWs and HCWs may allow scope for serving more families than in the pilot, retention and equity are important considerations.

Stipends are fixed amounts and thus do not account for varying delivery costs across communities. Turnover of CHWs is high, which may reflect issues with motivation that at least partly relate to compensation. Sustaining the value of the initial training investments relies on retaining these workers in the programme.

What are the next steps?

The next phase of this study will involve a final costing of the parenting intervention after its full implementation period, capturing the complete resource use and updating both the bottom-up and top-down estimates. This is expected to inform any cost-effectiveness analyses on the programme and provide high-level recommendations for future investment and potential scale-up within national systems.

Introduction

About this report

Thrive is supporting an early childhood development (ECD) research project in Tanzania called Kizazi Kijacho (KK, ‘the next generation’). One component of the research design is a randomised control trial (RCT) of a parenting and unconditional cash transfer (UCT) intervention in the Dodoma region.

This report presents estimates of the cost of the KK programme interventions from start-up (March 2021) to the end of its first phase when the target children are 1 year old on average (February 2024).

The findings show the costs for the three main treatment arms (parenting, UCT and combined parenting/UCT), as well as other types of breakdown, to help understand the main cost drivers. The resulting per child costs of the KK pilot interventions are not generalisable to a full implementation scenario and thus should not be taken to represent potential costs at scale. Different delivery models, operating structures, and economies of scale are some of the key reasons why pilot costs typically differ from costs at scale. While this report focuses on reporting the KK pilot costs, it also presents some simple scenario analysis of potential implementation costs under the NGO-delivery model used in the pilot.

This is the first of two reports. This first report covers the costs of the NGO-supported KK model that is being implemented on a pilot scale as a part of the RCT until the end of Phase 1, while the second will estimate costs of the full programme implemented up until the end of Phase 2.

The main audience for this first costing report are the NGOs that are managing the implementation of KK (Elizabeth Glaser Pediatric AIDS Foundation (EGPAF) and D-tree International (D-tree)), the KK research team, and KK steering committee, which includes senior government representatives. The NGOs will be interested to see the full picture of costs from their implementation efforts, and to draw lessons for future resource allocation for these types of intervention. The KK research team’s main interest, at this stage, will be looking at the costing results in the light of their analysis of the impact of the programme on ECD outcomes after the first phase of implementation. The KK steering committee will be keen to see the emerging evidence on the costs and impacts of the NGO-supported pilot as part of considerations for replication and scaling up under a government delivery model. After finalising this first report, the next step will be to consult stakeholders on feasible assumptions for government-delivery models of KK interventions (or parts of interventions) that are showing promise. D-tree will be particularly helpful in this regard as, in the last year of the programme, it coordinated an engagement process with government managers and health workers from the KK programme to work on strategies for sustainability.

About the Kizazi Kijacho programme

The KK RCT comprises a control arm and three treatment arms: (i) a parenting intervention; (ii) a UCT; and (iii) a combined parenting /UCT intervention. The research design includes an impact evaluation that aims to measure the relative effects of the different treatment arms on child development outcomes.

The study area is 387 communities in all 8 district councils in Dodoma region. The target population is women who were 5–7 months pregnant at baseline.⁵ The target sample size is 3,588 families (containing the targeted women and their newborn children). The study is first randomised at health dispensary (HD) level into the control and treatment groups, followed by a second community-level randomisation process within the treatment groups.⁶ After the babies were born, the target population for KK services became the primary caregiver and the child.⁷

The parenting intervention is being delivered by existing volunteer⁸ community health workers (CHWs) via an enhanced home visit programme and new group parenting sessions. Both CHWs and their supervisors – government healthcare workers (HCWs) – have been trained to use a digital app. As mentioned above, EGPAF and D-tree are supporting the interventions. D-tree is responsible for support related to the digital app, while EGPAF has a wider implementation role. The UCT is being delivered to caregivers by EGPAF directly via mobile money transfer. In the control areas, in principle, CHWs deliver the government's prescribed home visit service which covers health and nutrition but not wider parenting skills. The government's Tanzania Social Action Fund (TASAF) delivers cash transfers in both the treatment and the control areas to poor and targeted food-insecure households (~15% of all households in Tanzania).⁹

The KK design phase started in 2021, followed by training of CHWs and HCWs, which began in September 2022. Full implementation, meaning that all CHWs had started visiting KK families, began in January 2023 and continued for 14 months up to February 2024 for all 3 treatment arms.¹⁰ In parallel, the KK impact

⁵ In practice, the eligibility gestational age interval was increased to between 20 and 33 weeks of pregnancy (4.6–7.7 months) because there would not have been enough pregnant women otherwise.

⁶ At the second stage of randomisation, communities within first stage treatment groups (parenting or UCT) were randomised into a second layer of treatment. Parenting was split into parenting only and combined parenting/UCT. UCT was split into a fixed UCT and a varying UCT.

⁷ This is usually the biological mother of the child but not always.

⁸ CHWs are not on the government payroll. They sometimes receive stipends from projects or as part of specific government campaigns or short-term initiatives. The government has signalled its intention to pay regular stipends to CHWs.

⁹ See Almas et al. (2023). *Kizazi Kijacho: A randomised controlled trial of parenting and unconditional cash transfers in Tanzania. Baseline report*, THRIVE WORKING PAPER 1, Thrive, Oxford Policy Management, <https://thriveworkingpaper.org/resource-centre/kizazi-kijacho-trial-of-parenting-and-unconditional-cash-transfers-in-tanzania/>

¹⁰ CHWs were organised into three batches with training and implementation staggered. Batch 1 started implementing in November 2022, batch 2 from December 2022 and from January 2023, all 3 batches were implementing the parenting programme. The duration of

evaluation team conducted a baseline survey in October 2022 (Almas et. al., 2023) and an end-of-Phase 1 survey from February 2024 so it could estimate programme impacts when the target children were on average 1 year old. The KK parenting intervention (but not the UCT) has continued and is due to end in March 2025 after 27 months of implementation and when the target children are on average 2 years old (end of Phase 2). **The period covered by this costing study is the start of the design phase to the end of the first phase of the parenting programme and the endline point of the UCT programme in February 2024.** There will be a parenting programme end-of-Phase 2 impact evaluation survey from March 2025, and an accompanying Phase 2 costing and scaling study.

implementation is fixed at 27 months and so the batches will finish between January and March 2025.

Methods

This is a cost-economy study that estimates total programme costs, costs by treatment arm, costs per child for each arm, and costs by source (who paid for or donated the resources). Given that the impact of KK is being measured via an RCT that will estimate incremental impact compared to a counterfactual (i.e. the government's usual service for the target group), the cost study also estimates incremental costs (i.e. the additional costs of the intervention compared to the cost of the existing services).

Perspective

The study takes a societal perspective on costs. This means that it attempts to capture all the main resources required to make the interventions work (i.e. the primary costs) regardless of who pays for them or if they are provided in-kind (for example, the time spent by caregivers). Put another way, the aim is to quantify the cost of replicating the intervention to get the same outcomes that the impact evaluation finds. To keep the study manageable and to fix ideas, it has not attempted to estimate induced costs, which are the changes in resource use associated with the mediators of an intervention (i.e. knock-on costs arising from behaviour change after an intervention has been implemented).

Research questions

For each of the three KK intervention packages (parenting, UCT and combined parenting/UCT), the research questions that this first report responds to are:

- What is the cost of the resources needed to deliver the intervention under the NGO-supported delivery model? And what is the recurrent cost per child, after the investment costs are annualised?¹¹
- What are the main cost components and their cost drivers?
- How is the cost burden distributed among sources?
- What is the incremental cost of the intervention compared to the counterfactual (i.e. the cost of relevant services being delivered in the control group areas)?

¹¹ Annualisation is the process of converting large, one-time capital expenditures into an equivalent uniform annual amount throughout the duration of its lifespan. It is a useful way of comparing investments that have differing lifespans. It includes expenditure on things like IT equipment, software development and one-off training.

These research questions are a subset of the full list set out in the Kizazi Kijacho Costing Study: Design and Planning Report (Thrive 2023, Table 1, p.5). As noted earlier, the second costing report will address the outstanding questions related to replicating and scaling the interventions through government systems.

The research questions above have been applied to the NGO-supported delivery model operating at the pilot scale i.e. serving a target number of KK families/children, resulting in estimates of the cost per target child. Beyond this, still with a focus on the NGO-supported model, the report also examines costs per child under some scaled-up scenarios based on assumptions (e.g. about the capacity of CHWs to serve more than the target number of families).

Data sources

The study draws on two main sources of financial data: expenditure statements from EGPAF and D-tree, and a field survey specifically designed for this study. Caregiver numbers and estimates of the dosage of services come from the programme's monitoring data, which is particularly rich in the case of the parenting intervention because CHWs record data in an app in real time. Table 2 summarises the main categories of information needed against the main data sources.

Table 2: Data sources

	Training and delivery of interventions			Design and management	
	CGs, CHWs, HCWs #s	Dosage (quantity)	Unit costs (prices)	Design and start-up costs	Direct and overhead management costs
NGO budgets/expenditure statements			X	X	X
NGO finance systems			X		
Field survey		X	X		
KK parenting monitoring data [†]	X	X			
KK UCT monitoring data	X	X	X		
KK impact evaluation data		X	X		
Other secondary sources ^{††}			X		

Notes: † Data from the parenting app used by CHWs and from supervision forms filled by HCWs. †† For example, government salary scales, Bank of Tanzania exchange rate and inflation data.

Field survey details

There were two main reasons for carrying out a small-scale field survey, to complement the other sources of data.

- Reality check vs design: to observe how resources are being used in practice to deliver the interventions compared to the intended design. For example, some of the provided resources may not be used or may be used for different purposes.
- Resources not paid for by NGOs: to capture additional resources required to deliver the interventions that are not included in NGO financial statements. This includes the time of the CHWs and beneficiaries (since this has an opportunity cost),¹² the time of HCWs whose salaries are paid by the government, and any donated resources.¹³

Four interviewers collected data over 5 days in April 2024 from a case study sample of 40 respondents (8 CHWs, 8 HCWs and 24 caregivers). The sampling strategy aimed to capture the range of costs needed to deliver KK, and thus deliberately selected HD areas likely to be low and high cost. The survey took place in Chemba District Council (rural) and Dodoma City Council (urban), which have diverse contextual characteristics relevant to costs. The team visited eight HDs covering all study arms in each district. The actual sample size was the same as intended although there were some replacements made via a pre-designed protocol. Appendix A contains a summary of the sampling strategy. Table 3 shows the sample outturn by the type of intervention-related questions (some respondents answered more than one type).¹⁴

Table 3: Sample outturn – number of respondents by type of questions answered

Type of intervention-related questions	HCWs	CHWs	CGs	All
Parenting questions	4	4	12	20
Parenting control questions	4	4	12	20
UCT questions	4	4	12	20

Source: Kizazi Kijacho Costing Study: Fieldwork and Data Report (Thrive 2024)

In addition to the field survey in HD areas, the study team interviewed NGO staff and government officials responsible for managing or overseeing the KK programme. The number of interviews by organisation is in Table 4.

¹² The programme pays the CHWs (who are volunteers) a stipend to compensate them for their time and transport costs. Part of the reason for conducting the field survey was to collect information on CHW time use and transport costs to estimate if the value of the stipend covers this or if CHWs are also donating resources (time or out-of-pocket expenses for transport) to the programme.

¹³ A donated resource is a contribution to the programme that is not remunerated or reimbursed.

¹⁴ For example, caregivers in the combined parenting/UCT study arm were able to answer questions about parenting and UCT.

Table 4: Interviewers with management staff and government officials overseeing Kizazi Kijacho

Organisation	Location	Roles of people interviewed
EGPAF	Dar es Salaam	Senior manager ECD; 2 finance officers
EGPAF	Dodoma	2 project officers (parenting); finance officer (UCT); technical manager
D-tree	Dar es Salaam	Country director; deputy country director; engineering manager; senior technical advisor
Government	Dodoma	Regional CHW coordinator; Chamwino District CHW coordinator

Source: KK costing study team

Analysis methods

Costing methods

This study uses two costing methods:

- the ingredients method – bottom-up costing that involves listing all resources used in an intervention and then multiplying unit prices by quantities of resources used¹⁵
- a top-down method that uses apportionment rates to share costs that cannot be directly assigned between parenting and UCT treatment arms, and between different cost categories.¹⁶

The ingredients method has been used to cost the training of CHWs and HCWs, and the direct delivery of services (parenting and/or UCT) to caregivers. One key advantage of this method is that it makes price sensitivity analysis and estimating the costs of alternative scenarios straightforward – that is, it is relatively easy to look at the effect of changes in prices or dosage on total programme costs.

To help ensure the accuracy of the bottom-up ingredients methods costing, one key step in the analysis was to compare the resulting estimates with relevant total spending figures extracted from the NGO expenditure statements. This was only possible for resources that were funded by NGOs, and the comparisons are only approximate (since there is not a 1:1 mapping between the line items in NGO budgets and the training and delivery activities).

Programme design and start-up costs, and (nearly all) programme management costs have been estimated using the top-down apportionment approach. It is not practical to list the quantities and prices of all resources used for these activities (including items such as office rent and utilities) separately for parenting and the UCT. It is clear from the NGO expenditure statements which line items cover

¹⁵ Many references exist on the ingredients method of costing. See for example: Levin et al. (2018).

¹⁶ Apportionment is a common method of distributing overheads between different projects run by an organisation. The idea is to identify a rational method ('apportionment basis') of sharing costs that gives an approximate but fair distribution, for example by using relative staff numbers or floorspace or total project value.

these types of costs but not how to split them between parenting and UCT. By working with EGPAF,¹⁷ we developed two sets of apportionment rates, one based on the approximate level of effort of staff working on parenting and UCT activities during different periods of the programme, and another based on travel activity that was needed at different times for various programme management activities. Appendix B contains a list of line items drawn from the EGPAF financial statements that cover these design, start-up and management costs together with the apportionment rates and assumptions.

There is a degree of uncertainty about some of the unit costs, quantities and apportionment rates used to estimate the costs of the KK pilot programme. Particularly for large cost components, the assumptions can have a material effect on the resulting estimates. To explore this, the cost analysis includes sensitivity testing for some of the key cost components showing the effects of changing assumptions on the implementation costs per child.

Classification of costs

This study uses the classification of costs from the Childhood Cost Calculator (C3 tool).¹⁸ This recently launched tool is designed to capture costs of different types of ECD interventions in a standardised format, and to develop a global database of ECD programme costs. In line with C3, the costs of the KK programme have been tagged according to five cost categories and five resource types. General category definitions are given below. Details of how the categories apply to the KK programme costs are in the next chapter.

- Cost categories: programme design (all costs prior to the start of initial training);¹⁹ indirect management (overheads, administration and support function costs); training (of front-line delivery staff); direct delivery (services delivered to beneficiaries); and direct programme management (supervision and management of frontline delivery staff).
- Resource types: labour (all personnel costs); materials and equipment (all tangible items including communication); travel and accommodation; facilities (costs of physical spaces used by the programme, such as rental); and 'other' (e.g. cash transfers).

¹⁷ Based on interviews with EGPAF staff, and information from the KK Annual Report 2022–23 (EGPAF 2023), the study team compiled a draft set of apportionment rates. These were then discussed with EGPAF, with some adjustments made, before being finalised.

¹⁸ For further details on the system for classifying costs, see: Center for Universal Education at Brookings. (n.d.). *Childhood Cost Calculator (C3) user guide: A costing tool for education and early childhood development*. The Brookings Institution. <https://childhoodcostdata.org/cost-calculator/about-calculator/>.

¹⁹ Programme design took place between March 2021 and August 2022. Initial training of CHWs and HCWs happened between September and December 2022.

Other methodological issues

Investment vs recurrent costs

Each cost has been tagged according to whether it is an investment cost (capital resource that lasts more than one year) or a recurrent cost (used up within a year). Rather than including total investment costs in the year that they were paid for, these costs have been spread across their expected lifetime using a standard approach to annualisation.²⁰ This involves assumptions about interest rates and how many years the asset is expected to last.

For example, Table 5 below breaks down Phase 1 training components into either an investment or a recurrent cost. The investment components include all one-off training to HCW and CHWs where new content was delivered. These include the initial parenting training (Sep–Nov 2022) and the in-community practice session (Oct–Dec 2022), both assumed to generate benefits over the entire duration of the programme (2.25 years). Similarly, it also treats the group session training for parents under one year old (Aug 2023) as an investment with a shorter assumed lifespan (of 1.33 years). The parenting training for children aged 1–2 (Jan–Feb 2024) is similarly categorised as an investment, with benefits assumed to last one year. In contrast, the refresher training after the in-community practice (Oct–Dec 2022) and the Year 1 refresher training (April 2023) are classified as recurrent costs. For these recurrent components, lifespan and interest rate assumptions are marked as not applicable, since they do not create multi-year benefits and would need to be repeated.

Table 5: Categorisation of training by investment or recurrent and corresponding assumptions for Phase 1

Training	Duration per batch	Period held	Cost type	Assumed lifespan	Annual interest rate [†]
1. Initial parenting training	9 days (residential)	Sep–Nov 2022	Investment	2.25 years (Oct 2022– Dec 2024)	3%
2. In-community practice session	4 days (in-community)	Oct–Dec 2022	Investment	2.25 years (Oct 2022– Dec 2024)	3%
3. Refresher training after in-community practice	2 days (residential)	Oct–Dec 2022	Recurrent	n.a.	n.a.
4. Year 1 refresher training	2 days (residential)	April 2023	Recurrent	n.a.	n.a.
5. Group session training (for <1 yr olds)	3 days (residential)	Aug 2023	Investment	1.33 years (Sep 2023– Dec 2024)	3%
6. Parenting training (for 1–2 yr olds)	4 days (residential)	Jan–Feb 2024	Investment	1 year (Jan–Dec 2024)	n.a. ^{††}

Notes: † Annual interest rates account for the opportunity cost of annualization i.e. it is an equivalent interest that could be earned had it not been locked into the investment. This allows for a fair comparison of annualised costs when comparing investments that have varying lifespans. †† Since this training has only a

²⁰ See for example pp. 86-90 in Levin et al. (2018).

one-year lifespan, it is not annualised. However, only 1/6th of this cost is accounted for in the Phase 1 costs (i.e. for 2 out of 12 applicable months).

Inflation and exchange rates

All costs were originally expressed in nominal Tanzanian shillings (TZS), these were then adjusted for inflation²¹ and expressed in constant 2023 TZS prices and then converted to USD using an exchange rate of 1 USD = TZS 2,445.²²

Key assumptions and limitations

Exclusion of costs

The following are not captured: any programme design costs that were incurred by others apart from the NGOs; and research and evaluation costs such as time spent by the KK research team.

Uncertainty in estimates of NGO management costs

The distribution of EGPAF staff's time and level of travel between the parenting and UCT interventions (the basis for apportioning management costs) is based on key informant views, not a quantitative data source such as time-sheet analysis. It is also likely that some of the NGO management costs incurred in Phase 1 relate to planning for Phase 2 and thus overestimate Phase 1 costs.

Dosage of parenting intervention across arms

The study assumes that caregivers in the two parenting arms (parenting and parenting/UCT) get the same dosage of home visits and group sessions on average from the CHWs assigned to their arms. The CHWs are not involved in the UCT so it seems reasonable to assume that their services are not affected by whether the caregiver gets a UCT or not. This is supported by the KK Phase 1 evaluation results which found no significant differences in the frequency of home visits or group sessions between the arms (publication forthcoming).

Accuracy of dosage estimates from digital app data

Some CHWs have faced challenges in using the digital app to record their visits to families, and thus the dosage estimates for home visits may be underestimated. Unless this distortion is large, the knock-on effect on the cost estimates will be minimal because the value of CHW time in delivering home visits has been estimated as the actual stipends paid to them during implementation. There is evidence that the CHW stipend is more than sufficient to cover the time spent on home visits and group sessions at the recorded dosage (see **Box 1**). There would be additional donated time from caregivers as well as time spent by

²¹ Using the consumer prices index data from the Bank of Tanzania.

²² The USD: TZS exchange rate is the average of 12 monthly exchange rates for 2023 (EGPAF monthly financial statements and the Bank of Tanzania).

HCWs on supervision of any additional visits, but the results tables show that these costs account for a very small proportion of total costs.

Field survey results are indicative not representative

The small purposive sample was designed to give indicative results of the range of potential costs incurred by CHWs, HCWs and caregivers, not to provide estimates that are representative of the population of these groups in the KK programme. The time and costs reported in the survey were included directly in the bottom-up costing where they passed a 'sense-checking' test, based on the team's observations in the field and discussions with the KK research team. This was the case for the median time, travel costs and duration of visits/sessions reported by the sample of caregivers. The CHW and HCW estimates of travel time and costs were upper-bound estimates and so these were adjusted downwards to a more realistic level.

'As it costs approach'

The study has sought to capture all costs incurred in preparing for and delivering the KK programme up to February 2024. Given that this is a pilot that had not been tested in Tanzania before, this includes potential inefficiencies in implementation (e.g. using a more expensive training venue initially) so the resulting estimates most likely overestimate the cost of replicating the interventions using the same model at the same pilot scale.

Implementation dosage and input costs

This costing study covers the period from the start of the KK programme design in 2021 to February 2024 (end of Phase 1), which includes an average of 15 months of implementation. The start of implementation was staggered across 3 months (November 2022 to January 2023) resulting in 14 months of full implementation from January 2023 when all CHWs had started working.

The sections below set out the intended design together with data on the actual implementation of the parenting intervention and the UCT. This includes the nature and intensity of services delivered to caregivers and the key personnel delivering them.

Parenting implementation

Starting with the parenting intervention, Table 6, Table 7 and Table 8 summarise the intended design, split between the two relevant study arms, and the actual dosage received by the caregivers over the 14-month Phase 1 period.

Approximate estimates of the government's existing home visit services being delivered in the control areas are in the right-hand column of the table for comparison.

Table 6: Summary of parenting intervention and information on control group

Study arm	Parenting	Parenting and UCT	Control
# targeted CGs	761	726	856
#registered CGs	690	663	n.a. [†]
Drop from target to registration (%)	-9%	-9%	n.a.
#CHWs	77	77	n.a.
#HCWs	46	42	n.a.
Description of intervention for Phase 1 Control services (per govt protocol)	5 residential training sessions for CHWs/HCWs ~19 CHW home visits guided by a digital app ^{††} 6 group parenting sessions, monthly from Sep 2023 Monthly supervision visits from HCWs		~ 12 CHW home visits focused on health and nutrition support ^{†††}

Notes: † n.a. means not applicable. †† The start of implementation was staggered across three batches of CHWs, which means that the expected number of home visits as per KK protocol by February 2024 ranged between 18 and 20. ††† KK home visits include support for parenting practices while this is not part of government home visit materials for CHWs.

Table 7: Summary of actual dosage of intervention by parenting intervention

Study arm	Parenting	Parenting + UCT	Control
Average # home visits per CG	10.6 home visits		~ 0.6 to 1.4 home visits
Average duration of visit	~60 mins		not available
Average # group sessions per CHW	6 group sessions		n.a.
Average duration of group session	~110 mins		n.a.
Shares of available group sessions attended by CGs	60%		n.a.
Average # HCW supervision visits per CHW month	0.3 supervision visits		n.a.

Notes: † For details on the source of dosage estimates, see Appendix D.

Table 8: Dropout and turnover during Phase 1

Study arm	Parenting	Parenting + UCT	Control
#CGs still registered at end of Phase 1	616	588	n.a.
CG dropout from registration to end of Phase 1	-11%	-11%	n.a.
Turnover of CHWs (%)	19%		n.a.
Turnover of HCWs (%)	21%		n.a.

Notes: † CHWs and HCWs who leave the programme were replaced as soon as practicable.

Design vs actual dosage

Compared with the intended KK parenting design dosage, CHWs have delivered fewer home visits than anticipated but far more than appear to be happening in the control areas (which in any case do not formally cover support for parenting). It is important to note that:

- the three antenatal care visits in the KK protocol were not always applicable given birth timing
- granular visit-by-visit analysis reveals that compliance was particularly low for the three expected visits immediately after delivery (Day 0, Day 3 and Day 8) and, in hindsight, the design may not be well-suited to context.²³

CHWs held group parenting sessions as planned but caregivers only attended 60% of sessions available to them, possibly due to distance and related costs. HCW supervision of CHW visits is far lower than the protocol. Whereas the protocol prescribed one HCW visit per month per CHW, in reality CHW only

²³ The KK protocol for the first three CHW visits after birth follows the government protocol, but discussions with key informants about the KK findings reveal that visits immediately following birth may go against cultural norms.

received 0.3 HCW visits per month, or approximately one visit every three months. This is likely due to HCW time constraints.

Community health worker and healthcare worker training

In KK, the frontline delivery of the parenting intervention is carried out by CHWs. These were selected from the existing cadre of CHWs who are attached to government HDs. CHWs are unpaid volunteers, although when they participate in projects, such as KK, the government requires them to be given stipends and/or allowances to cover their costs at set rates. Under KK, a group of 154 CHWs (organised into 3 batches, split roughly evenly into about 50 per batch) from 8 district councils in the Dodoma region, received 5 separate residential trainings starting in September 2022. EGPAF engaged a small group of 3 to 4 government officers to facilitate the training:

- first training (Sept to Nov 2022) – 9 days followed by field practice in the CHW's home community; covered KK home visit protocol using the digital app, content focused on parenting children up to 12 months old
- refresher training (Oct to Dec 2022) – 2 days immediately after field practice before implementation
- refresher training 2 (Apr 2023) – 2 days
- group session training (Aug 2023) – 3 days; covered KK group session protocol for children 6 months and older
- second training (Jan to Feb 2024)– 4 days; content focused on parenting children 13–26 months old.

Home visits and group sessions

Following the training, the first batch of KK CHWs started implementing home visits from November 2022, the second and third batches of CHWs began their home visits in December 2022 and January 2023 respectively. This means that on average there was 15 months of implementation by February 2024,²⁴ with caregivers receiving an average of 10.6 home visits lasting about 60 minutes each. CHWs were equipped with smart phones so that they could use the KK digital app during their visits. CHWs spent approximately 30 minutes travelling to and from each home visit. Following the training on group parenting sessions, CHWs began holding these sessions monthly, either at their HD, a community venue or a caregiver's home. CHWs held an average of 6 group sessions lasting about 110 minutes each over the 6 months from September 2023 to February 2024. Caregivers spent an average of 40 minutes travelling to and from the location of each group session but they only attended 60% of available group sessions.

²⁴ The staggered implementation means the batches carried out 16, 15 and 14 months of implementation respectively by February 2024. The full implementation period will be 27 months for each batch, so there will also be a staggered ending with the third batch finishing in March 2025.

Supervision of community health workers

The supervision and management of KK CHWs is layered. HCWs are the immediate supervisors of CHWs. HCWs are qualified health care professionals who are employed and salaried by the government and based at HDs. One of their existing roles is to supervise CHWs, but the KK supervision role required additional activities. HCWs participated in the five training sessions listed above so they would be able to support and mentor CHWs to conduct home visits and group sessions. On average CHWs received one home visit supervision visit every three months from their HCW during home visits. HCWs spent up about 30 minutes travelling to and from households to conduct these supervision visits.²⁵ HCWs attended 30% of available groups sessions over the 6 months that these were implemented. This adds to the supervision that HCWs do of the home visits.

EGPAF employed two full-time project officers, based in Dodoma, to manage and supervise the parenting intervention. During implementation, about 50% of project officer time each month was spent visiting HCWs and CHWs to provide support, in months when training was not taking place. Their remaining time was largely spent on conducting phone-based supervision of CHWs carrying out home visits.²⁶ The project officers attended each of the KK training sessions full-time to support the facilitators.

A quarterly monitoring team, comprising two regional government officers,²⁷ two council government officers²⁸ and six EGPAF and D-tree staff members, make up the final layer of supervision support for CHWs and HCWs. This team carried out 4 supervision visits, each taking 5 days, during the 15 months. The team visited 1 or 2 councils each time, typically staying overnight, so that most of the eight councils in Dodoma were covered by February 2024. The regional CHW coordinator (RCHWCO) and the eight council CHW coordinators (CCHWCOs), who are part of quarterly monitoring, all participated in the five training sessions listed above.

Digital app and supervision dashboard

D-tree designed the digital app that CHWs use during home visits, as well as the online dashboard that HCWs and CCHWCOs access to monitor CHWs.²⁹ D-tree staff provided training on using the app and dashboard, as well as ongoing support during quarterly monitoring visits or at other times as needed.

²⁵ The estimate of travel time for HCWs was assumed to be the same as for CHWs.

²⁶ The information about the time-use of project officers is approximate. It comes from key informant interviews with the project officers and their manager.

²⁷ Regional CHW coordinator and the regional reproductive and child health coordinator.

²⁸ District CHW coordinator and the district reproductive and child health coordinator.

²⁹ The online dashboard provides supervisors with basic information on the frequency and timing of CHW home visits compared to the protocol. The programme provided HCWs with smart phones, but CCHWCOs were required to use their own smart phones to access the online platform.

UCT implementation

The KK UCT intervention is simpler by design than the parenting intervention. Table 9 summarises the basic design and the actual number of caregivers who were paid UCTs in the two relevant study arms. The table does not include the control arm because the Government's cash transfer programme (Productive Social Safety Net (PSSN) II managed by TASAF) operates across all the study arm locations, so the KK UCT is additional to the existing service in the treatment areas.

Table 9: Summary of Kizazi Kijacho UCT intervention

Study arm	UCT only	Parenting and UCT
Programme design		
# target CGs	1,239	726
# registered CGs	1,108	650
Drop target to registered (%)	-10.6%	-10.5%
Description of intervention	Registration of CGs, then cash transfers by mobile money every 2 months – 6 payments in total.	
	Each CG assigned into low (~USD 13), medium (~USD 32) or high payment (~USD 45) amount [†]	Each CG assigned into medium payment amount (~USD 32)
Actual dosage/CGs paid		
# payments per CG	6	6
# CGs paid	1,101	642
Registered CGs not paid (%)	0.6%	1.2%

Notes: † In Tanzanian shillings the amounts are: low (TZS 32,000), medium (TZS 77,000) and high (TZS 109,000).

Source: KK UCT monitoring data (EGPAF)

Rationale for the payment amounts

The middle amount (~USD 32) was estimated to be equivalent to the government's PSSN II payment to a mother with a newborn baby (KK's target group) every 2 months.³⁰ The low amount (~USD 13) was an initial ballpark estimate of the cost per family of KK's parenting intervention per 2 months, assuming that each CHW was responsible for 60 families. The high amount (~USD 45) is the sum of the low and middle amounts. The idea was to calibrate the amounts to be affordable to government and to facilitate comparisons between the study arms to see if barriers to improving ECD outcomes largely relate to finance or parenting information/support or both.

³⁰ For a PSSN II household that includes a mother with a newborn baby, the minimum and maximum payments every 2 months are: TZS 44,000 and TZS 111,000, which is an average of TZS 77,000 (~USD 32).

Design vs actual dosage

Only a handful of caregivers who registered for UCT did not receive a payment. And almost all the caregivers who were paid received the expected six payments.³¹

Registration

Following KK's baseline survey in October 2022, which identified pregnant women eligible for the UCT, EGPAF staff called the targeted women to invite them to participate in the UCT and to give details of the process and location for registration. A team from EGPAF spent the next 3–5 months travelling to HDs and to communities to register caregivers. This took longer, and was more complicated, than had been envisaged. EGPAF hired temporary staff to help with completing the registration exercise. About 11% of targeted women had not been registered by the end for various reasons.³²

Payments

EGPAF finance staff made the payments to caregivers via a mobile money system to their mobile phones. Caregivers received payments in Tanzanian shillings between March 2023 and February 2024.

Management and grievances

After the payments started, a small team of EGPAF staff managed the payments and the various issues that arose. Considerable routine manual record keeping and follow-up with caregivers was necessary after each payment. Challenges included: payments being returned because the caregiver's mobile phone was not active at the time of payment; incorrect phone numbers (a field verification exercise had to be conducted requiring caregivers to travel to their HD); difficulties contacting caregivers to check that payments had been made; and caregivers wanting to change phone numbers. An EGPAF staff phone number was available for caregivers to call if they had problems in receiving payments.

Kizazi Kijacho input unit costs

The resources used in the KK programme have been assigned to five main cost categories: three direct categories (i.e. costs closely related to delivering the intervention) and two indirect categories (overheads). The direct costs of the parenting intervention are briefly described below.

³¹ According to EGPAF only a handful of UCT recipients did not get all six payments.

³² The reasons are detailed in the EGPAF (2023) KK Annual Review Report 2022–23.

Training (direct cost)

The resources required to deliver the KK training sessions to CHWs and HCWs. This means all labour (NGO staff, government officials and CHW volunteer time),³³ training materials, travel and accommodation at the training venues, venue rental and food.

Direct programme delivery (direct cost)

From the start of implementation, the cost of labour for home visits and group sessions (CHW stipends and caregiver volunteer time), travel, as well as materials (smart phones, solar chargers, communication bundles, material kits).³⁴

Direct programme management (direct cost)

The costs incurred by HCWs, other government officials and NGO staff on supportive supervision during the programme. This means all labour costs, materials (smart phones, communication bundles for HCWs), travel and accommodation costs for field visits.

Turning to the direct costs for the UCT intervention, there was no substantive training of NGO staff to deliver the UCT and so the training category does not apply. The cash transfer payments themselves fall under direct programme delivery. NGO staff time, materials, travel and accommodation required for registration of caregivers, making payments, and following up on issues are direct programme management costs.

Drilling further down into costs,

³³ CHWs were not paid a stipend during training so the time they spent during training has been valued because this has an opportunity cost.

³⁴ CHWs are paid a stipend to cover their time and transport costs for carrying out home visits and group sessions. Our estimates show that the value of the stipend more than covered the time and transport costs incurred by CHWs in delivering the average dosage (see Box 1). In principle, if the stipend was insufficient to cover CHW costs then any additional time and costs would have been valued and included in the direct programme delivery category.

Table 10 contains the main unit costs driving the direct costs of the KK programme. The middle two columns show whether the cost is a donated resource (a contribution to the programme that is not remunerated or reimbursed), and whether it is an investment item (lasts more than 1 year).

Table 10: Kizazi Kijacho unit costs (direct costs)

Category	Donated resource	Investment resource	Recurrent resource	Unit	USD (2023 prices)
Parenting					
Labour					
CHW stipend			✓	USD/month-person	20
CHW volunteer time (during training) [†]	✓			USD/day	3.6
HCW salary			✓	USD/month-person	348
Regional/district govt official salary			✓	USD/month-person	489
CG time	✓			USD/day	1.3
Materials and equipment					
Smart phones for CHWs/HCWs		✓		USD/phone	164
Solar chargers for CHWs		✓		USD/charger	31
Comms bundle for CHWs/HCWs				USD/month-person	5
Material kit for CHWs ^{††}		✓		USD/kit	138
Server hosting			✓	USD/month	922
Travel and accommodation					
Training – DSA [‡]		✓ ^{‡‡}	✓ ^{‡‡}	USD/night-person	33–53
Training – conference package		✓ ^{‡‡}	✓ ^{‡‡}	USD/day-person	11
Training – transport		✓ ^{‡‡}	✓ ^{‡‡}	USD/return journey	9
Supervision – transport allowance for HCW			✓	USD/month-CHW	6
Quarterly supervision – DSA for govt officials			✓	USD/night-person	53
Facilities					
Training – venue hire		✓ ^{‡‡}	✓ ^{‡‡}	USD/night-person	111
Group session – room rental	✓			USD/day-person	0.4
UCT					
Cash transfer – low amount			✓	USD/2 months-person	13
Cash transfer – medium amount			✓	USD/2 months-person	32
Cash transfer – high amount			✓	USD/2 months-person	45

Notes: † CHWs were paid a stipend during implementation, which more than covered their time and transport cost (see Box 2 for analysis of this) but during training they were not given a stipend. †† The material kit supplied to CHWs comprised: KK t-shirt, KK bag, KK hat, KK guideline, KK flipbook for home visits, KK flipbook for group sessions, government flipbook/Bango Kitita. Some additional materials were supplied in the January/February 2024 training to support CHW work with children aged 1–2, including materials to make toys. The lower rate is for CHWs. The higher rate applies to HCWs and other government officials. ‡‡ Depending on the training involved, it would be an investment or recurrent resource. Refer to Table 5 for details.

Source: See Appendix D, which sets out the source of each unit cost and explanatory notes.

The remaining two cost categories are for programme overheads. Brief descriptions of the type of costs incurred under each are given below. These categories apply to both the parenting and the UCT interventions, with the relative shares determined by the approximate level of effort of NGO staff and the extent which activities required travel (the apportionment rates are in Appendix B).

Programme design costs (overhead)

All costs incurred during the 18-month design period from March 2021 to August 2022. This includes NGO staff time for design and development of the programme elements such as the digital app for CHWs, dashboard for supervisors, curriculum and material kit for CHWs. It also covers NGO administrative and office running costs as well as their fixed percentage overhead costs (~13-15% of total costs).

Indirect programme management (overhead)

From September 2022 onwards (start of training), this covers NGO administrative and office running costs (rental, supplies, phones etc.) as well as their fixed percentage overhead costs (~13-15% of total costs).

Table 11 further categorises the different cost categories plus transfers for the UCT arms into investment versus recurrent costs. Investment costs include programme design expenditures and durable inputs such as smartphones (for HCWs/CHWs) and parenting materials/manuals. Recurrent costs cover ongoing programme management (indirect and direct) and direct delivery expenses such as staff time, supervision/labour costs, travel/transport stipends, communication bundles, and transfers for the UCT arms.

Table 11: Summary of key activities categorised as investment or recurrent in Phase 1, by cost category

Cost category	Types of activities	Cost type	Lifespan
1. Programme design (both treatments)	NGO costs incurred on programme staff salaries, office supplies, travel etc. during design phase	Investment	4 or 3 years depending on whether incurred in 2021 or 2022 respectively
2. Indirect programme management (both treatments)	NGO costs on support staff salaries, office rental, and other fixed % charged to cover HQ costs	Recurrent	n/a
3. Direct programme management (both treatments)	NGO programme staff salaries, HCW labour costs for supervision, other govt. staff costs for supportive supervision	Recurrent	n/a
	Smart phones for HCWs	Investment	3 years
4. Direct delivery (parenting arms only)	Transport allowance/stipends for CHWs and HCWs, communication bundles for CHWs	Recurrent	n/a
	Smartphones and chargers for CHWs, all parenting materials (such as manuals etc.)	Investment	3 years (IT equipment) 2 years (manuals)
5. Training (parenting arms only)	DSA, Transport allowance, conference package, labour costs of facilitations, CHWs, HCWs, and govt. officials	Investment or Recurrent depending on training	As in Table 5 above
6. Transfers (UCT arms only)	Unconditional cash transfers	Recurrent	n/a

Results

Programme costs

The costing analysis of the KK interventions gives insight into the resources that were required to implement these up to February 2024 ('Phase 1'). These resources include both provider costs and donated costs.

- Provider costs – refer to the financial costs incurred by service providers in delivering the intervention. These include both direct costs (such as for training, materials, and CHW/HCW stipends) as well as indirect costs (such as administration and operational expenses as well as pre-intervention design). On KK, the service providers incurring financial costs are the NGOs (EGPAF and D-tree) and the government.
- Donated costs – since we are using a societal perspective on costs, we also estimate the value of time and resources donated towards the implementation of this programme, reflecting the mostly non-financial inputs essential for programme delivery.

The provider costs will be of particular interest to those interested in budgeting for a similar pilot programme, while understanding donated costs is relevant to governments in making choices about investments that consider the welfare of society overall. Getting a full picture of programme costs is critical for assessing the cost-effectiveness of the pilot interventions, and for giving insights into the sustainability and scalability of the intervention.

Another key distinction that we make in the results tables that follow is between investment and recurrent costs. Investment costs are one-off or upfront expenditures that generate benefits over multiple years, whereas recurrent costs are ongoing expenditures that must be incurred repeatedly to run the programme.

This section presents detailed breakdowns of programme costs, discusses the main cost drivers, how these costs are distributed across different sources and estimates the incremental cost of the intervention (with respect to the control group).

Which interventions are the least and most costly?

To estimate the total cost of the programme, we separately calculated the costs of the parenting intervention and the UCT intervention in constant 2023 US dollars. We then apportioned total costs to the different treatment arms based on the target number of caregivers in each.³⁵ Table 12 compares the total programme costs as well as the investment and recurrent costs of each treatment arm per child.

Table 12: Programme costs per child by treatment arm (constant 2023 USD)

Per child cost estimate	Parenting only ^{††}	Parenting and UCT ^{††}	UCT only – high	UCT only – medium	UCT only – low
Total programme (total cost per targeted child[†])	832.39	1079.26	319.60	252.08	146.87
- o/w Investment costs	278.62	280.12	1.50	1.50	1.50
- o/w Recurrent costs	541.90	786.90	317.52	250.00	144.79
- o/w Donated costs [‡]	11.87	12.23	0.58	0.58	0.58
Total recurrent Provider cost (per child per month (pcpm))^{‡‡}	38.71	56.21	22.68⁵	17.86	10.34
Total recurrent programme cost (per child per month (pcpm) – includes donated costs)^{‡‡}	39.19	56.72	22.72	17.90	10.38
Number of targeted caregivers	761	726	921	166	151

Notes: † 'Per child' here refers to the cost per originally targeted family/caregiver under KK's design and does not represent actual children reached in the programme. †† Cost per child for parenting arms reported may be a conservative estimate because of potential spillover effects on other families in the community. ‡ These are resources donated by CHWs and caregivers as part of participation in the programme and can be either investment or recurrent depending on the activity involved ‡‡ Recurrent costs per month are calculated as recurrent costs divided by total months until the end of Phase 1 (i.e. 14 months) and give an estimate of implementation costs to run the programme at current dosage and scale. (5) Costs per targeted child in 'UCT only – high' arm were found to be lower than the UCT entitlement per month (USD 23.20) because the actual amount paid out was less due to 11% attrition of beneficiaries compared to the targeted sample. Although similar attrition was also observed in the 'UCT only – medium' arm (11%) and 'UCT only – low' arm (12%), the lower entitlement amount and fewer targeted caregivers in these arms meant cost per targeted child did not go below the entitled monthly amount (USD 16.44 and USD 7.03 respectively).

³⁵ We used the original target number of caregivers rather than the actual number of recipients for two reasons: (i) to maintain comparability with the research's intention-to-treat impact estimates; and (ii) because all design considerations were based on these target figures. The exception was the UCT transfers where we used actual expenditures incurred in each treatment arm, as these could be precisely determined.

Source: Authors' calculations

Comparing across the treatment arms, the UCT only arms have far lower per child costs compared to the parenting arms. It is worth noting though that the parenting intervention's per child costs assume CHWs provided the home visits and group parenting sessions only to the targeted caregivers. In practice, they were expected to continue providing their routine services to other families as well, and it is plausible that the improved capacities of the CHWs from the programme had spillover effects on these additional families, which are not captured by the programme.³⁶

The combined parenting/UCT arm is 30% more expensive per child than the parenting only arm, and this difference rises to 45% when we only look at the recurrent costs.

It is notable that the investment costs for the UCT arms represent a relatively small share of their total costs compared with the distribution seen in the parenting interventions. This reflects the relative simplicity of the design of the UCT compared with the parenting intervention where investment costs include a bespoke app and training materials and design.

Nearly all the programme's costs are incurred by the providers. Donated costs comprise only a small portion of the total programme costs (1.4% in parenting only, 1.09% in parenting and UCT and less than 1% in UCT only arms).

What are the main cost components of the programme?

Understanding the main cost components for the programme and their underlying drivers helps in identifying which components contribute most to overall costs and can help both optimise resource allocation and identify where efficiencies can be gained. The following section examines both investment and recurrent costs – by their different cost categories and resource types – to help understand the main cost drivers and their implications for programme implementation.

Parenting costs — by cost category

We first examine the costs of the parenting activities across its two treatment arms – parenting only and combined parenting/UCT – by its five key cost categories. These are outlined below in

³⁶ Following government guidance for CHW workload, the KK research team had estimated that CHWs would continue to provide services to approximately 60 families (compared to the 9–10 families per CHW targeted by the KK programme), albeit not using the app and including children up to 5 years old. No data is available to estimate the extent to which this assumption held, but the low level of home visits in the control group, and views from the NGO staff, suggest that minimal additional home visits were made to non-KK families.

Table 13 and Table 14 respectively. Detailed break-up by resource type of all five key cost categories are shown in Table C.3 in Appendix C.

Table 13: Programme costs of parenting only intervention by cost category (constant 2023 USD)

Category	Investment Total	Recurrent Total	Both Total	Share of total provider costs (%)
Total provider cost	212,031.23	412,384.25	624,415.48	100%
Overhead	109,643.47	142,136.26	251,779.74	40%
– Indirect management	-	142,127.49	142,127.49	23%
– Programme design	109,636.71	-	109,636.71	18%
Direct	102,387.76	270,247.98	327,635.74	60%
– Training	85,603.75	106,606.23	192,209.98	31%
– Direct delivery	13,290.56	36,560.69	49,851.26	8%
– Direct programme management	3,943.45	127,081.06	130,574.50	21%
Total provider cost per child	278.62	541.90	820.52	
Total provider cost of training per child			252.58	
Total provider cost per month per child	19.90	38.71	58.61	
Total donated costs	3,836.29	5,196.95	9,033.25	1%
- CHW training time	3,836.29	2,248.25	6,085.14	
- Caregiver time and resources	-	2,611.52	2,611.52	
- Group session facilities (rental)	-	336.58	336.58	
Total donated cost per child			11.37	
Total donated cost per month per child	0.36	0.49	0.85	
Total overall programme costs	215,867.528	417,581.20	633,448.72	
- Total cost per child	283.66	548.73	832.39	
- Total cost per month per child	20.26	39.19	59.46	

Table 14: Programme costs of the combined parenting and UCT intervention by cost category (constant 2023 USD)

Category	Investment Total	Recurrent Total	Both Total	Share of total provider costs (%)
Total provider cost	203,370.47	571,209.04	774,660.51	100%
Overhead	105,691.74	170,686.70	276,378.44	36%
– Indirect management	-	170,686.70	170,686.70	22%
– Programme design	105,691.74	-	105,691.74	14%
Direct	97,678.73	400,603.34	498,282.07	64%
– Training	81,666.65	101,703.19	183,369.83	24%
– Direct delivery	12,679.30	34,879.19	47,558.49	6%
– Direct programme management	3,322.77	137,323.32	140,656.09	18%
– Transfers	-	126,697.65	126,697.65	16%
Total provider cost per child	280.12	786.90	1,067.03	
Total provider cost of training per child			252.58	
Total provider cost per month per child	20.01	56.21	76.22	
Total donated costs	3,659.85	5,222.25	8,882.11	1%
– CHW training time	3,659.85	2,145.42	5,805.27	

- Caregiver time and resources	-	2,491.41	2,491.41	
- Group session facilities (rental)	-	321.10	321.10	
- Caregiver time (during UCT registration)	-	264.32	264.32	
Total donated cost per child	- -	-	12.23	-
Total donated cost per month per child	0.36	0.51	0.87	
Total overall programme costs	207,030.32	576,512.29	783,542.61	
- Total cost per child	285.17	794.09	1,079.26	
- Total cost of implementation per month per child	20.37	56.72	77.09	

We find the following major cost components of the parenting programme.

Training

As can be seen in

Table 13 and Table 14, training was the most significant cost component of the KK programme,³⁷ mainly driven by its high intensity, extended duration, and the large number of participants involved. Across 5 rounds, the training sessions spanned a total of 20 days and were conducted as residential programmes for all 154 CHWs, 88 HCWs and government RCHWCOs and CCHWCOs.

The largest contributor to training costs was participation allowances, which accounted for 70% of total training costs. These included daily subsistence allowances (DSAs) to cover food and accommodation, calculated based on participant seniority, as well as fixed transport allowances determined by participants' locations. Additionally, per participant costs for conference packages – including food, stationery, and training materials – further added to this expenditure.

Labour costs also represented a significant share (28%) of training costs, covering the value of time contributed by government officials and trainers. These were estimated based on per day participation costs linked to salary scales for HCWs and local government officials, alongside trainer costs for facilitators from the implementing NGOs.

Overhead costs

Overhead costs were another large component of overall programme costs,³⁸ with a larger share in indirect management costs.

Indirect management accounted for 22–23% of total provider costs in both parenting treatment arms,³⁹ primarily driven by indirect costs charged by the implementing NGOs to cover non-programmatic expenses (64% of indirect management costs).⁴⁰ Additionally, it included the costs of administrative staff from the NGOs that supported programme implementation and a share of their overall cross-programmatic overheads such as rent and utilities (21% of indirect management costs).

Programme design costs made up 18% of total provider costs in the parenting-only arm and 14% in the parenting and UCT arm. The largest contributor to these costs was labour, NGO staff and their consultants, accounting for 74% of total programme design costs. The remaining share was largely driven by other fixed indirect costs during design – including VAT, import duties, and other charges (15% of programme design costs).

Other provider costs

The remaining provider costs of the parenting intervention in the two parenting treatment arms are distributed across direct programme management and direct delivery costs.

³⁷ 31% of total provider costs in parenting only arm and 24% of total provider costs in combined parenting/UCT arm.

³⁸ 40% of total provider costs in parenting only arm and 364% of total provider costs in the combined parenting/UCT arm.

³⁹ The indirect management cost in the combined parenting/UCT arm includes the indirect management cost of the UCT arm also (and not just parenting).

⁴⁰ The NGOs charged an additional fixed indirect cost of 15%–18% on the total cost incurred by them on the programme, mainly to cover their headquarter costs.

Direct programme management accounted for the larger share of these other provider costs.⁴¹ The predominant driver of these costs was labour, and transport expenses associated with providing direct supervisory support (88% of direct programme management costs), primarily incurred by HCWs as well as EGPAF and D-tree staff. The remaining 12% covered materials and equipment necessary for programme management, including the cost of smartphones and chargers for HCWs and server hosting fees for the KK app.

Direct delivery costs contributed the smallest share of overall programme expenditure, ranging from 6%–8%. Most of these costs (64%) were stipends paid to CHWs and travel allowances for HCWs involved in programme delivery. The remaining 37% comprised expenditures on materials and equipment for implementation, including smartphones, accessories and communication bundles for CHWs, as well as learning materials used for home visits and group parenting sessions.

Donated resources

These were a minor part of the parenting programme and were equivalent to around 1% of the provider costs. The main contributor of this was the time cost of CHWs for their training participation (nearly 67% of donated resources).⁴² The other contributors to this were caregiver time and resources donated to participate in KK activities (29%)⁴³ and notional rental cost of facilities used for hosting the group parenting session (4%).

UCT costs — by cost category

In addition to the costs associated with the parenting intervention, the UCT treatment arms (UCT only – three payment amounts; and parenting and UCT – medium payment amount) have their own cost components, which are outlined here. The detailed cost table of the UCT only intervention is shown in Table 15.

Table 15: Total programme cost of UCT only intervention by cost category (constant 2023 USD)

Category	Investment	Recurrent	Both	Share of total provider costs (%)
	Total	Total	Total	
Total provider cost	1,860.42	355,804.09	357,664.51	100%
Overhead	1,860.42	59,832.54	61,692.96	17%
– Indirect management	-	59,832.54	59,832.54	17%
– Programme design	1,860.42	-	1,860.42	1%
Direct	-	295,971.55	295,971.55	83%
– Training	-	-	-	0%
– Direct delivery	-	-	-	0%

⁴¹ 21% of total provider costs in the parenting-only arm and 18% in the combined parenting/UCT arm.

⁴² We valued the CHW time based using their reported median wage rate on similar activities, as reported in *Kizazi Kijacho Impact Evaluation Baseline Report*, (Almas et. al., 2023).

⁴³ We considered caregiver time as the opportunity cost of participation using a relevant housing and budgeting survey, as per methodology outlined in Lopez et al. (2021). Caregiver resource value was estimated from primary data. We valued caregiver time using the median female wage rates for agricultural workers in Dodoma district based on the Integrated Labour Force Survey 2020/21 (National Bureau of Statistics 2022).

– Direct programme management	-	27,432.08	27,432.08	8%
– Transfers	-	268,539.47	268,539.47	75%
Total provider cost per child	1.50	287.40	288.91	
Total provider cost per month per child	0.11	20.53	20.64	
Total donated costs	-	450.72	450.72	0.1%
– Caregiver time (during UCT registration)	-	450.72	450.72	
Total donated cost per child per month			0.36	
Total donated cost per month per child		0.03	0.03	
Total overall programme costs	1,860.42	356,254.81	358,115.23	
Total cost per child	1.50	287.77	289.27	
Total cost per month per child	0.11	20.55	20.66	

Note: This table represents the aggregate costs of all three UCT arms only.

Transfers

The direct costs of cash transfers were the largest cost component in the UCT interventions.⁴⁴ These costs primarily consisted of the amounts disbursed to beneficiaries as part of the cash transfer programme by the managing NGO. In addition to the transfer amounts, an extra transaction fee of TZS 200 per recipient (USD 0.08)⁴⁵ and cash withdrawal charges ranging from 4%–7% of the transferred amount were also included.⁴⁶

Indirect programme management

The next largest share of costs in the UCT interventions were indirect programme management costs incurred by the NGO.⁴⁷ As explained above, these were the fixed cost charged on all NGO programme costs for non-programmatic expenditure along with some administrative support staff time and resources charged to the programme.

Other provider costs

Unlike parenting, the UCT intervention did not incur any large direct costs on training and had a smaller programme design cost (1% of total provider cost). The only other substantial cost incurred was for direct programme management,

⁴⁴ On average it accounted for 75% of overall provider costs of UCT-only arms and 16% of the parenting and UCT arm. However, as seen above, the latter's costs were mainly driven by the parenting intervention.

⁴⁵ This was to cover mobile money transfer transaction costs incurred by the NGO.

⁴⁶ During field surveys, caregivers reported receiving an additional amount to cover cash withdrawal charges from their mobile wallets. However, we were not able to confirm this from the NGO expense records. This was an additional 4% of the amount for medium and high UCT arms and 7% for the low UCT arm. We used the average withdrawal charges levied by the three largest mobile wallet providers in Tanzania in estimating these charges.

⁴⁷ It accounted for, on average, 17% of overall provider costs of the UCT-only arm.

mainly attributed to NGO staff time on the registration and management of the UCT programme (8% of total provider cost).

Donated resources

The proportion of donated resources for UCT was a small fraction of the cost of the total provider cost⁴⁸ and was mainly incurred on the time spent by recipients on registering for the UCT with the NGO. We have assumed that the NGOs incurred the withdrawal charges required for recipients to access their transferred amount as cash from their mobile money wallets. If this is incorrect, and this cost fell on the recipients, then the approximate donated amounts would be 4% (for medium and high UCT transfers) or 7% (for low UCT transfers) of the provider cost on transfers (the exact amounts depend on the mobile money service provider). This is further examined in 'Sensitivity analysis'.

⁴⁸ 0.1% of total provider cost, on average, for the UCT-only arm, and 1% in the combined parenting/UCT arm.

How is the cost burden distributed among sources?

The costs of the KK intervention were shared across multiple stakeholders, including NGOs, the government, and other contributors. Each entity played a distinct role in providing resources for various aspects of the programme, from direct service delivery to administrative and overhead expenses. Table 16 and Table 17 below outline the distribution of costs among these sources, highlighting their respective contributions.

Table 16: Total programme cost of parenting (both arms) by source (constant 2023 USD)

Cost source/ paying agent	Investment	Recurrent	Both	Share of total provider costs (%)
	Total	Total	Total	
Total provider cost	414,310.70	805,802.07	1,220,112.77	100%
Overhead costs	214,244.21	277,735.38	491,979.59	40%
- NGOs	214,244.21	277,735.38	491,979.59	40%
Direct costs	200,066.48	528,573.37	728,639.85	60%
- NGOs	176,901.20	507,639.71	684,540.91	56%
- Government	23,165.28	20,426.98	43,592.26	4%
Total donated costs	7,496.15	10,154.89	17,651.04	1%
- CHWs	7,496.15	4,394.27	11,890.42	
- CG	-	5,102.93	5,102.93	
- Other (group session facility rentals)	-	657.69	657.69	
Total programme costs	421,806.84	815,956.96	1,237,763.80	
- NGOs	391,145.41	785,375.09	1,176,520.50	
- Government	23,165.28	20,426.98	43,592.26	
- Others donated	7,496.15	10,154.89	17,651.04	

Note: The parenting costs in the parenting-only and combined parenting/UCT arm have been aggregated in this table.

Table 17: Total programme cost of UCT (all arms) by source (constant 2023 USD)

Cost source/ paying agent	Investment Total	Recurrent Total	Both Total	Share of total provider costs (%)
Total provider costs	2,951.42	533,676.30	536,627.72	100%
Overhead costs	2,951.42	94,920.12	97,871.55	18%
- NGOs	2,951.42	94,920.12	97,871.55	18%
Direct costs	-	438,756.18	438,756.18	82%
- NGOs	-	438,756.18	438,756.18	82%
Total donated costs	-	715.04	715.04	0.1%
- Caregivers	-	715.04	715.04	
Total programme costs	2,951.42	534,391.34	537,342.76	
- NGOs	2,951.42	533,676.30	536,627.72	
- Donated	-	715.04	715.04	

Note: The UCT costs in the parenting/UCT arm and in the 3 UCT only arms have been aggregated in this table.

As seen above, 97% of the parenting intervention's costs and 100% of the UCT intervention's costs were borne by the NGOs implementing the programme. Most of the remainder of the parenting costs were covered by the government (3%), through the labour contribution of its officials (namely, HCWs, council and regional RCH and CHW officials) as part of their supervisory role in the programme. Donations account for close to 2% of parenting costs but only 0.1% of UCT costs.⁴⁹ Some points to note in these cost distributions.

Community health worker contributions

As part of the parenting intervention, CHWs are paid TZS 50,000 each month as a stipend. This was expected to cover their labour (i.e. the value of their time) as well as transport and other costs incurred to deliver the home visits and group sessions. We found this stipend to adequately cover their implementation costs during Phase 1 (see Box 1). Thus, CHWs are not donating resources during programme delivery given the actual dosage of services delivered. However, during training, CHW allowances only covered the material cost of participation (i.e. accommodation, transport and subsistence), not the time that they spent during training, which is thus considered a donated resource.

Healthcare worker contributions

HCWs are given a transport allowance of TZS 15,000 per month for every CHW they supervise in the programme. When assessing the adequacy of this allowance at current intervention dosage levels, we find it is sufficient to cover their transport costs (see Box 1). Unlike CHWs, HCWs are full-time government employees, and their time is valued as a provider cost incurred by the

⁴⁹ This figure was obtained by aggregating UCT costs over all four study arms. The share of donated resources in total costs differs slightly across the UCT-only study arms, but all are <1%.

government. This also applies to district and regional government officials who participate in quarterly supportive supervisions of the parenting intervention.

Excluded costs

In the costing field survey, some CHWs and HCWs also reported donating resources towards parenting activities (e.g. materials to create toys and providing paper for drawings). Given the small number of respondents who reported donating resources,⁵⁰ we have excluded the value of these donated resources in our cost estimates (time and resources donated by caregivers have been included in our estimates of parenting).

Box 1: Adequacy of allowances in the parenting programme

Do the stipends/allowances paid to CHWs and HCWs adequately cover costs?

To assess the adequacy of stipends/allowances provided to CHWs and HCWs for delivering the KK intervention, we conducted a bottom-up cost estimation of their labour and transport expenses. We estimated the unit time required for each activity — such as home visits, group parenting sessions and supervision — from our field survey. A programme-level time use estimate was obtained by multiplying these unit times with current dosage levels, as observed in KK digital app data. We then applied standardised per-hour labour costs, reflecting government salary scales for HCWs and prevailing compensation benchmarks for CHWs to get labour costs. Additionally, we calculated transport costs based on the number of visits made (from KK digital app data) with average per trip transport costs (from the field survey). This approach provided an indicative estimate of the actual costs incurred by CHWs and HCWs in fulfilling their programme responsibilities. We then compared these estimated costs against the stipends currently provided to assess whether they adequately compensate CHWs and HCWs for their efforts. If the stipends fall significantly short of the estimated costs, it may indicate a risk of under compensation, which could affect motivation and retention, potentially impacting programme effectiveness. Conversely, if stipends exceed estimated costs, this suggests a buffer that may support additional responsibilities or allow for programme adjustments. By examining this alignment, we can determine whether current stipend levels are sufficient to sustain engagement at the required intensity. (See Table C.4 in Appendix C for further details).

CHW stipends

Using median CHW wage rates obtained from the baseline KK impact evaluation survey, our bottom-up estimates of labour costs for the current dosage levels as of February 2024 are USD 15,265 while transport costs are USD 11,674 — that is, a total cost of CHW direct delivery of approximately USD 26,939. The total stipends paid out during the implementation period are approximately USD 47,293. Thus, the current dosage level costs approximately 57% of the stipend payments, which suggests buffer room for expansion. However, it must be noted that this gap is mainly because the number of CHW visits are below that intended in the design (as per KK digital app data). Preliminary findings from the KK process evaluation suggest that CHWs may not have consistently recorded all activities on the app due to technical issues faced while using it (process

⁵⁰ Only three of eight CHWs and one of eight HCWs reported contributing their own resources. These were also not of substantive value (under TZS 10,000 or USD 4 in total per respondent).

evaluation — forthcoming). Thus, any increase in dosage needs to carefully consider actual CHW workloads on the programme.

HCW transport allowance

As per current dosage levels of supervisory support provided by HCWs, the bottom-up estimates of transport costs to February 2024 are approximately USD 1,619. However, travel allowances paid out during the implementation period totalled USD 14,188 — that is, about 11% of transport allowances have been used in HCW travel requirements for supportive supervision. This suggests there is room to substantially increase supervisory visits without affecting transport costs. However, just like with CHW activities, the current intensity of visits recorded on the digital app is below that intended by programme design, some of which might be attributable to inadequate use of the app (process evaluation publication — forthcoming). Moreover, HCWs may have also incurred transport costs related to KK which are not recorded on the app (e.g. to help troubleshoot app issues or sort out any stipend payment issues.)

Note: Although the comparisons above suggests scope to increase dosage at least to some extent without the need to pay additional stipends or transport allowances, multiple other factors (such as motivation and time availability) and costs (such as on direct programme management) would ultimately determine whether dosage could be increased or not. This is in addition to the reliability concerns of KK digital app data to determine current dosage.

What is the incremental cost of the intervention?

In this section we estimate the incremental cost of the KK intervention – that is, determining the additional resources required to deliver KK beyond existing services. This applies to the parenting intervention because the government already runs a CHW home visit service, which KK builds on in the treatment areas, but not to the UCT intervention because the government’s cash transfer programme is implemented in both treatment and control areas. The reason for computing incremental costs is so that they can later be related to the incremental impact estimates that the KK research team is working on.

To estimate the incremental costs of KK parenting, we assess the extent of parenting services already occurring in control areas and compare them with similar services undertaken in treatment areas. By analysing the difference in service provision intensity between these two settings, we can estimate the additional costs incurred due to the intervention. Through this approach we attempt to isolate the programme’s financial burden from routine health system activities.

The KK parenting intervention includes home visits, group sessions and HCW supervision of these sessions. The Phase 1 evaluation survey data suggests there is some level of antenatal and postnatal care home visits undertaken by CHWs in control areas, but the other parts of the KK intervention are not part of the government’s programme and hence, it seems reasonable to assume that these are not present in control areas.⁵¹ Thus, to calculate the incremental cost of the intervention, we deduct the costs of home visits occurring in control from the cost of delivering home visits in treatment.

⁵¹ Such as group parenting sessions and HCW supervision of home visits.

Table 18 shows the difference in average home visit intensity between treatment and control areas.

Table 18: Comparison of home visits intensity – treatment and control

Home-visit type	Treatment	Control	Notes on control estimates
Antenatal visits	0.7	0.4	20% of control group caregivers reported receiving at least one visit by CHW at home during pregnancy. Those receiving at least one visit received two home visits on average.
Postnatal visits	9.9	0.2	10% of control group caregivers reported receiving at least one home visit by a CHW. Those who received postnatal care visits received two visits on average.
Child visits		0–0.8	Information on child visits in control areas is not available from the Phase 1 survey. We assume 20% of caregivers (same coverage as for antenatal) got the full four follow-up child visits as per government schedule - as an upper bound (and none as a lower bound).
Total	10.6	0.6–1.4	Gives a range of lower and upper bound control cost estimates.

Source: KK Phase 1 Impact Evaluation findings; authors' assumptions

To be conservative, we have assumed that the value of CHW time and travel costs per home visit (as covered by the KK stipend) are the same in the treatment and control areas. This means that the relative cost of CHW time and travel in treatment and control areas simply reflects the ratio of home visit intensity shown in the table above. Note that this may overestimate costs in the control areas if, for example, home visits are shorter than in the treatment areas. Step-by-step control estimates calculation methodology is shown in Appendix C.

Using this method of obtaining control estimates, we get the lower and upper bound incremental costs for implementation (per child per month) for each intervention arm, as shown in Table 19.

Note that in these calculations we have excluded all other costs associated with CHWs in the control areas such as management and other overhead costs because it seems reasonable to assume that these costs continue to apply to the CHWs in treatment areas too. The costing field survey findings support the assumption that the KK CHWs continue to work at the HD delivering their 'business-as-usual' service to a wider group of families and to participate in health campaigns and other one-off initiatives. Put another way, the assumption is that NGO management/overhead costs in running KK do not substitute for the government's management/overhead costs in treatment areas and are additional.⁵²

⁵² It is possible that the HCW's supervision time for KK CHWs might be substituting for some business-as-usual supervision time of these CHWs in treatment areas. Also, the time spent by HCWs in attending KK training may also be substituting time from their routine support to CHWs.

Table 19: Incremental recurrent cost estimation, by intervention arm (constant 2023 USD)

Cost item	Parenting	Parenting and UCT	UCT only [†]
Total programme recurrent costs	417,581.20	576,512.29	356,254.81
o/w Provider costs	412,384.25	571,290.04	355,804.09
o/w Donated resources	5,196.95	5,222.25	450.72
CHW total stipend for time and travel	24,203.11	23,089.95	-
o/w Stipends – home visits only	22,853.73	21,802.64	-
Control area costs of CHW time and travel to home visits			
Lower bound	1,293.61	1,234.11	-
Upper bound	3,018.42	2,879.59	-
Incremental recurrent costs			
Lower bound	396,201.08	555,943.76	356,254.81
Upper bound	397,745.98	557,589.24	356,254.81
Per-child incremental recurrent cost, per month			
Lower bound	37.17	54.70	20.55 [†]
Upper bound	37.33	54.86	20.55 [†]

Note: [†] Control costs for UCT are zero because the UCT programme was over and above any existing cash transfer programme occurring in treatment areas – that is, even if a caregiver was receiving the government's social protection cash transfer, they would receive the UCT also. We also assume any fixed costs for UCT management would be similar in both treatment and control.

Given the cost structure of the KK parenting programme, together with the relatively low level of home visits in control areas, it is not surprising to find that the incremental costs of implementation are very close to the total implementation costs in treatment areas. Direct delivery costs (where the home visits costs fall) only accounts for 12% of provider implementation costs in the parenting arm, so there would have been a much higher level of home visit delivery in the control areas to drive a material difference between incremental and total costs.

Robustness of pilot cost estimates and scaling scenarios

This section presents two complementary analyses that help assess the robustness and practical implications of the costing study findings.

- The sensitivity analysis explores how variations in key costing assumptions affect the overall provider cost estimates for the KK programme. This focuses on cost components that are most influential and where there is relatively more uncertainty in the estimates.
- The scaling analysis assesses the feasibility of delivering the KK interventions at a larger scale, assuming the same NGO-delivery model at current dosage levels. It looks specifically at how many families CHWs could reach if the interventions were to be delivered within the limits of their existing stipends or the time they are willing to commit to the programme.

Together, these analyses provide insights into the cost-efficiency of the pilot and its potential scalability, noting that the analysis is confined to the NGO-supported model. The second KK report in this series will explore the potential costs associated with scaling KK via a government delivery model where cost structures are likely to differ.

Sensitivity analysis

In this study we have used many assumptions in estimating programme costs. To account for uncertainty in the key assumptions and to improve the robustness of our findings, we undertook a sensitivity analysis. This exercise tests how variations in key input parameters affect overall cost estimates. It helps to identify which cost components have the greatest influence on total programme costs and assesses the stability of conclusions under different assumptions. Given that some of these assumptions include both investment and recurrent costs, we have reported the sensitivity analysis results in terms of total provider costs (in per child per month terms).

Parenting — sensitivity analysis

We have undertaken a sensitivity analysis of three cost categories of the parenting intervention. The results are summarised in Table 20. Against each cost category, we have highlighted the current assumption and outlined alternate assumptions to provide a lower and upper bound estimate of its respective per child per month cost.

Table 20: Summary of parenting sensitivity analysis (all costs in constant 2023 USD)

Cost category	Total provider cost (pcpm)	Current assumption	Alternate assumptions	Lower	Upper
Direct programme management	12.26	NGO (EGPAF) staff time distribution – 30% time in Jan–Jun 23; 50% in July–Dec 23; 30% in Jan–Feb 24	Lower –half the time distribution Upper –double the time distribution	10.82 (↓12%)	15.12 (↑23%)
Training	18.04	DSA rates for participation * CHW = TZS 80,000 per day * HCW = TZS 130,000 per day * Other govt = TZS 130,000 per day	Lower –TZS 90,000 per day for all (as per original budget) Upper –TZS 130,000 per day for all (most govt staff's DSA rate)	17.04 (↓6%)	20.73 (↑15%)
	18.04	All other training costs – inflation-adjusted estimates from 2021 budgeted values (transport, venue hire, conference package etc.)	Lower – reduced by 25% Upper – increased by 25%	17.50 (↓3%)	18.59 (↑3%)

Indirect programme management	13.34	Indirect costs apportionment – apportioned to parenting at 90% during design (Sep 21–Aug22) and 50%–77% during training + implementation (Sep 22–Feb24)	<i>Lower</i> – all apportionment reduced to 50% throughout <i>Upper</i> - training + implementation apportionment increased to 90% throughout	11.02 (↓17%)	16.13 (↑21%)
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Some inferences from the table above.

- Direct programme management is fairly sensitive to uncertainty in apportionment rates for staff costs. The current assumption is that the implementing NGO used staff time on the parenting and UCT interventions in proportion to what they set out in their original budget (which was then adjusted based on the NGO programme manager's inputs). When this assumption was varied to some alternatives (i.e. halving or doubling the proportion of time allocated to parenting), it led to a 12% decrease or 23% increase in costs or a net 35% variance. This reflects the substantial role of NGO staff time in managing and supervising the programme, suggesting that even small changes in workload distribution can meaningfully impact overall direct management costs.
- Training costs are moderately sensitive to uncertainty in DSA rates. Using a uniform intermediate DSA rate reduced training costs by 6%,⁵³ while a uniform higher rate increased them by 15%.⁵⁴ All other training-related costs (e.g. transport, venue and conference package) showed less variability. For example, even with a 25% adjustment up or down in other training inputs, total training costs varied by only ±3%. This reinforces that participation allowances are a key cost driver in training and any changes to the training structure would significantly influence its total costs.⁵⁵
- Indirect programme management costs are quite sensitive to apportionment rates between parenting and UCT interventions. We had assumed indirect programme management costs are apportioned between the parenting and UCT interventions in proportion to the overall costs incurred by the implementing NGOs on each arm as per their original budget. Changes in apportionment rates for indirect costs resulted in a cost swing of ±17%–21%, indicating some sensitivity and the roughly linearly varying nature of these costs. It also underscores the importance of clear and consistent cost-sharing arrangements in joint programmes.

⁵³ The original NGO budget had used a uniform rate of TZS 90,000 for every training participant (which is higher than CHW rate but lower than HCW and other government officials' rates).

⁵⁴ We used a higher rate of TZS 130,000 which is the most common DSA rate for most junior health staff. In case CHWs are mainstreamed into the permanent healthcare workforce, many of them may fall under this DSA category.

⁵⁵ For example, shifting from a residential training to non-residential training will halve DSA rates for participation.

In conclusion, these findings suggest that the current cost estimates for the parenting intervention are reasonably robust, with some specific areas such as staff time allocation and training DSAs being more sensitive to changes. It also offers potential areas for improving cost-efficiency or requiring greater cost scrutiny in future budgeting and planning of parenting interventions.

UCT — sensitivity analysis

Unlike the parenting intervention, the costs for the largest cost drivers were known with reasonable certainty in the UCT intervention. However, as above, the influence of staff time allocation for direct and indirect programme management costs is equally relevant for UCT. The sensitivity analysis presented in Table 21 instead focuses on assumptions about withdrawal charges as these are not negligible costs relative to the UCT amounts and it is not clear who bore these costs.

Table 21: Summary of UCT sensitivity analysis (all costs in constant 2023 USD)

Transfers	Total provider recurrent cost (pcpm) [†]	Current assumption	Sensitivity variations	Lower	Upper
Low	10.34	Withdrawal charges – we assume the NGOs bore the beneficiary cost of withdrawing UCT amount as cash from mobile wallet	Lower – NGO does not bear withdrawal charges (borne by caregivers) Upper – current scenario	10.00 (↓3%)	10.34
Medium	17.86			17.34 (↓3%)	17.86
High	22.68			22.00 (↓3%)	22.68

Note: † The provider costs per targeted child for transfers is below the entitlement monthly amounts high (USD 23.20) arm because of (i) attrition between targeting and registration of caregivers; and because (ii) the total costs have been distributed over the 14-month intervention period (as opposed to the entitlement of 6 bi-monthly payments i.e. meant to be for 12 months).

The sensitivity analysis of assumptions about withdrawal costs for the UCT arm reveals that the direct transfer costs are relatively stable across all three transfer levels—low, medium, and high.

Currently, it is assumed that NGOs cover these withdrawal fees.⁵⁶ If instead the caregivers bear these charges, the estimated provider cost decreases modestly by 3%. This low level of sensitivity suggests that most of the cost variation across UCT arms is driven by the actual transfer amount.

However, transferring the responsibility of withdrawal charges to caregivers significantly increases their cost burden – from USD 0.03 to USD 0.64 per child per month, an over 20-fold increase. While this may seem small in absolute terms, it represents a substantial share of total costs borne by caregivers, with low-income households particularly negatively affected.

⁵⁶ In our field survey, caregivers said they received the additional withdrawal charges as part of their cash transfer amount. However, a sample transaction report from the NGO did not have these withdrawal charges although the programme manager also seemed to think these withdrawal charges were borne by the NGO.

Scaling the current NGO-supported model via improvements in cost-efficiency

As part of the KK costing study, we conducted a preliminary scaling analysis to explore how the programme might scale up within the existing NGO-supported structures, while maintaining its current dosage levels for both parenting and UCT. This analysis focused on understanding the potential reach within the existing resource and time constraints.

Scaling up parenting

For the parenting intervention, we examined how many families each CHW could feasibly reach under two scenarios: one where delivery costs match the value of the stipend paid to CHWs, and another where their current workload on KK matches their self-reported availability for CHW duties.

CHW delivery costs match stipends paid

In this analysis, we used bottom-up estimates of labour and transport costs incurred by CHWs to carry out home visits and group parenting sessions on KK (referred to as 'bottom-up CHW cost') and compared it to the stipends they had been paid (referred to as 'provider CHW cost'). As per data from the KK app, 154 CHWs are currently expected to be undertaking parenting interventions with at most 1,353 caregivers who were eligible and registered for the programme across both the parenting and combined parenting/UCT arms – that is, serving 8.79 families per CHW at dosage levels recorded in the KK app. The bottom-up costing of this dosage and reach roughly translates to provider costs of USD 1.42 per child per month (constant 2023 USD) during the 14 months of full implementation. CHWs were paid a stipend of TZS 50,000 per month, which translates to USD 2.49 per child per month (constant 2023 USD). We varied the number of caregivers that CHWs could cover to match the stipends they are paid, as shown in Table 22. We find that at current dosage levels, the CHWs could cover about 16 caregivers instead of the current approximately 9 families (a 78% increase) for the current stipend amounts paid using the current programme structure.

Table 22: Scaling scenarios – CHW delivery costs vs stipends paid (constant 2023 USD)

CGs per CHW	Total CG	Coverage change	Provider CHW cost (pcpm)	Bottom-up CHW cost (pcpm)	Provider recurrent cost (pcpm)	Comments
4.83	744	-50%	4.53	1.49	77.41	
7.82	1,204	-19%	2.80	1.43	47.81	Registered and still on the app at end of Phase 1
8.32	1,281	-14%	2.63	1.43	44.93	Registered caregivers used in PE analysis
8.79	1,353	-9%	2.49	1.42	38.71	Families registered on app
9.66	1,487	0%	2.26	1.42	37.37	Targeted by design
10	1,540	4%	2.19	1.41	37.37	
15.76	2,427	63%	1.39	1.39	23.71	Equilibrium where CHW bottom-up costs match stipends paid
20	3,080	107%	1.09	1.38	18.69	

29.56	4,552	206%	0.74	1.37	12.64	Families theoretically covered based on time CHWs claim could be devoted per week for CHW work
30	4,620	211%	0.73	1.37	12.46	
40	6,160	314%	0.55	1.36	9.34	
50	7,700	418%	0.44	1.36	7.47	
60	9,240	521%	0.36	1.35	6.23	Maximum families per CHW as per KK design

Source: Authors' calculations

CHW workloads match their self-reported estimates

In this analysis, we estimated the total time CHWs spent on KK parenting activities (home visits and group sessions), as well as their routine work at health facilities, by using field survey data on self-reported time-use, at current dosage levels (as reported by the KK app). The data implies that **CHWs spent on average, 12.4 hours per week on CHW-related work**, including KK (3.5 hours per week for KK activities and 8.9 hours per week for routine work at health facilities). Another data source, the Phase 1 evaluation survey, found that CHWs were **willing to work for 19.0 hours per week on CHW-related work**. Table 23 shows how many caregivers CHWs could serve at the current dosage of home visits and group session, if they were to work for more hours (or for fewer hours). We find that if they were to work 19 hours, as per their self-reported limit, they could serve about 30 families.

Table 23: Scaling scenarios – CHW workload vs self-reported time available for CHW work

Total CGs	CGs per CHW	Home visits per month	Group sessions per month	HV time (hrs/month)*	Time for GS (hrs/month)†	Other time spent (hrs/month)††	Total workload (hrs/week)	Comments
744	4.83	3.66	1	6.88	3.33	40	11.16	
1,204	7.82	5.92	1	11.15	3.33	40	12.11	Registered CGs still on app at end of Phase 1
1,281	8.32	6.30	1	11.86	3.33	40	12.27	Registered CGs used in PE analysis
1,353	8.79	6.66	1	12.53	3.33	40	12.41	CGs registered on app
1,487	9.66	7.31	1	13.77	3.33	40	12.69	Targeted by design
1,540	10	7.57	1	14.26	3.33	40	12.80	
3,080	20	15.14	1	28.52	3.33	40	15.97	
4,552	29.56	22.38	1	42.15	3.33	40	19.00	Maximum CGs theoretically possible based on self-reported time available for CHW work (Equilibrium)
4,620	30	22.71	1	42.78	3.33	40	19.14	
6,160	40	30.29	2	57.04	6.67	40	23.05	
7,700	50	37.86	2	71.30	6.67	40	26.21	
9,240	60	45.43	2	85.56	6.67	40	29.38	Max families per CHW as per KK design

Note: * Home visit time calculated for current dosage (10 visits in 14 months). † Group sessions time calculated for current dosage (one session per month). We have assumed a group session would be for a maximum of 30 caregivers at a time. †† Other time spent is the median time CHWs reported spending on routine activities at the health facility in the costing field survey.

Source: Authors' calculations

We also estimated provider costs if the parenting only and parenting and UCT arms were scaled up at current dosage. It was found that if CHWs were to visit 30 caregivers (maximum possible as per self-reported time available), the provider recurrent cost for the parenting only arm was USD 12.64 while for the parenting and UCT arm was USD 28.91. Further details on all relevant scaling scenarios are outlined in Appendix E.

Scaling up UCT

In the UCT arms, we varied the rate of caregiver attrition (i.e. the difference between targeting and registering caregivers) to assess how the cash transfer costs would change. Thus, it is strictly not a scaling up assessment but rather a cost-efficiency assessment of the cash transfer component. Currently, all three UCT arms have caregiver attrition of 11%–12% compared to the target as per the design. Varying the attrition rates, only affects some of the cost components. Direct management and overhead costs ('other costs') for the UCT have been apportioned between the three payment arms using the number of targeted caregivers in each arm. This means that changing the attrition rate does not change the value of 'other costs' per target child. What does change as the attrition rate is varied is the cost of transfers per target child. These cost efficiency scenarios are outlined for each UCT arm in Table 24, Table 25, and Table 26.. As expected, lowering the attrition rate results in the transfer cost approaching actual entitlement amounts per child.

Table 24: Scaling scenarios for UCT – low with varying attrition (constant 2023 USD)

Targeted CG	Attrition %	Actual CG paid	Transfers cost (pcpm) [†]	Other costs (pcpm)	Total cost (pcpm)
151	0%	151	6.02	5.14	11.17
151	5%	143	5.72	5.14	10.87
151	10%	136	5.42	5.14	10.57
151*	11.92%	133	5.31	5.14	10.45
151	15%	128	5.12	5.14	10.26
151	20%	121	4.82	5.14	9.96

Note: * Row in bold represents current scenario. † The transfer cost even at 0% attrition is less than the monthly entitlement amount (USD 7.02) because these costs have been calculated over the entire implementation period of 14 months while caregivers received 6 bi-monthly payments i.e. covering 12 months.

Table 25: Scaling scenarios for UCT – medium with varying attrition (constant 2023 USD)

Targeted CG	Attrition %	Actual CG paid	Transfers cost (pcpm) [†]	Other costs (pcpm)	Total cost (pcpm)
894	0%	894	14.10	5.14	19.24

894	5%	849	13.39	5.14	18.53
894	10%	805	12.69	5.14	17.83
894*	11.30%	793	12.53	5.14	17.67
894	15%	760	11.98	5.14	17.12
894	20%	715	11.27	5.14	16.42

Note: * Row in bold represents current scenario. † The transfer cost, even at 0% attrition, is less than the monthly entitlement amount (USD 16.44) because these costs have been calculated over the entire implementation period of 14 months while caregivers received six bi-monthly payments i.e. covering 12 months.

Table 26: Scaling scenarios for UCT – high with varying attrition (constant 2023 USD)

Targeted CG	Attrition %	Actual CG paid	Transfers cost (pcpm) [†]	Other costs (pcpm)	Total cost (pcpm)
922	0%	922	19.89	5.14	25.03
922	5%	876	18.90	5.14	24.04
922	10%	830	17.90	5.14	23.04
922*	11.39%	817	17.65	5.14	22.79
922	15%	784	16.91	5.14	22.05
922	20%	738	15.91	5.14	21.06

Note: * Row in bold represents current scenario. † The transfer cost even at 0% attrition is less than the monthly entitlement amount (USD 23.20) because these costs have been calculated over the entire implementation period of 14 months while caregivers received six bi-monthly payments i.e. covering 12 months.

Comparison with government ECD budgets

Estimates of overall ECD-sensitive⁵⁷ government spending in Tanzania provides important context for the cost estimates of the KK pilot programme produced in this report. By situating the programme's costs within the broader national spending landscape, we can get a sense of the resource envelope available for integrating the intervention into government systems at scale, and to inform the approach to working out a sustainable government delivery model for KK (the focus of the next KK costing report).

Before presenting the ECD-sensitive budget estimates, it is important to reiterate that the costs of an NGO-supported pilot are likely to be very different from when the government replicates or scales it up. It is likely that a government-delivery model of KK would have a different cost structure, intervention dosage, scale and

⁵⁷ UNICEF's definition of ECD-sensitive budgets was applied in this study: ECD-sensitive budgets cover interventions that are not explicitly designed to contribute to ECD outcomes but do so indirectly. For example, immunisation is important for child health and contributes to child development, but it is not explicitly designed to improve the social-emotional or cognitive development of a child (UNICEF 2023).

level of cost per child compared to the NGO-supported pilot. Box 2 provides further details on factors that drive these differences.

Box 2: Why pilot costs are not comparable with government implemented models

There are several reasons why the costs of the KK programme pilot are not directly comparable to a full-scale replication (at the current size) or scaling up by the government.

Different operating structures

The pilot was implemented by NGOs using dedicated project teams and support systems. During replication or scaling, delivery would shift to government infrastructure which have different cost structures and may benefit from existing institutional capacity. For example, the direct programme management costs would be very different if district CHW coordinators take on the delivery roles currently played by full-time, better-paid NGO staff.

Different delivery models

Replicating or scaling the KK programme through government systems would likely require design changes to make the intervention more affordable, feasible, and broadly acceptable. This could include adjustments to training modality and intensity, supervision frequency, or use of digital app, all of which could influence the cost structure.

Economies of scale

Many of the programme's recurrent costs, especially around programme management are fixed or semi-fixed (e.g. server hosting costs for the app). These costs were spread over a relatively small number of caregivers during the pilot, inflating per-child costs. A scaleup would allow these fixed costs to be distributed across a much larger population, reducing the cost per child significantly. Moreover, costs of printing and producing materials for the programme also reduce when procured at a larger scale, further reducing costs per child.

Ramping up effects

During the pilot, there is a ramp-up period where implementation systems are being set up (e.g. setting up the digital app and UCT payment infrastructure, various trainings etc.). Even with the support of high-capacity NGOs, it takes time for intervention to start operating smoothly. This means the programme may not be operating at full capacity for the implementation period, resulting in lower apparent efficiency. At scale, with mature systems and continuous enrolment, the intervention would be more likely to reach and maintain steady-state operations, improving cost-efficiency.

In summary, the pilot represents a high-intensity, resource-heavy delivery model designed for proof of concept and research. A full-scale government-led version would likely be leaner, benefit from existing systems and economies of scale, and be adapted for affordability and sustainability in the long term.

The Thrive team conducted a situation analysis of existing ECD planning and public finance systems in Tanzania in 2024. One of the objectives of this study was to estimate the total ECD-sensitive budget allocation in FY 2023/24 aligned with the five pillars of the Nurturing Care Framework for ECD.⁵⁸ The findings are summarised below in Table 27.

⁵⁸ This is a widely used global framework (World Health Organization (WHO), UNICEF and World Bank Group 2018) that is part of Tanzania's National Multisectoral Early Childhood

Table 27: The government's ECD-sensitive budget allocation in FY 2023/24

NCF pillar	Annual ECD-sensitive budget	Annual ECD-sensitive budget [†]	Annual per-child ECD budget allocation ^{††}
	FY 2023/24 (TZS billion)	FY 2023/24 (USD million)	FY 2023/24 (USD)
Health including nutrition and CHWs	846	346	21
o/w primary healthcare	370	151	9
Early learning	1,092	447	28
o/w pre-primary and primary schooling	962	394	24
Nutrition (other)	54	22	1
o/w agricultural services	53	22	1
Security and safety	412	169	10
o/w social protection (incl. cash transfers)	142.5	58	4
o/w water, sanitation, and hygiene	244	100	6
o/w other security and safety	26	11	1
Responsive caregiving^{†††}	Not estimated		
Total	2,405	984	61

Note: † USD estimates calculated from nominal Tanzania budget numbers for 2023 with exchange rates used in the current costing study for 2023 constant TZS prices i.e. 1 USD = TZS 2,445. † † Per child estimates calculated on the total ECD-relevant population aged 0–8 as per the 2022 census (16.2 million). † † † Responsive caregiving was excluded since this relates to the content of a service provided (such as building awareness on improved childcare through CHWs or health facilities) rather than a service delivery platform that could be identified through budget books. However, any budgets allocated toward it have probably been included in the health pillar.

Source: Nayanpally. et al. (2025)

The findings suggest that the total annual ECD-sensitive budget allocation for FY 2023/24 was approximately USD 61 per child or USD 5 per child per month. For the elements relevant to KK namely, CHWs and cash transfers, the annual per-child resource envelope was USD 9 and USD 4 respectively against the health pillar and social protection pillar respectively (or USD 0.75 and USD 0.33 per child per month respectively). These are likely to be conservative estimates but they give an indicative sense of the level of spend that new ECD initiatives would need to align with to be affordable for government.⁵⁹

As would be expected for the reasons outlined in Box 2, the recurrent cost per child per month of the KK pilot interventions to providers (USD 10.34 for the lowest cost arm in the KK pilot programme) are considerably higher than the government's ECD-sensitive spend per child per month of USD 5.

Development Programme (Ministry of Health, Community Development, Gender, Elderly and Children 2021).

⁵⁹ The study team was not able to review detailed activity-line budgets and used high-level apportionments on publicly available aggregated budget data instead. Thus, there may be scope for a revision of these estimates with more detailed budget data.

Discussion and next steps

Implications of the findings

This costing study has estimated total programme costs, costs by treatment arm, per child costs, and the distribution of costs by funding source for the KK pilot programme. Given that the impact of KK is being evaluated through an RCT, which measures incremental impact compared to the government's usual services, the costing study has also estimated incremental costs – the additional financial requirements beyond existing service provision. The study has taken a societal perspective, capturing all primary costs necessary for implementing the intervention, regardless of whether they are paid for or donated/provided in kind. The findings of the four research questions are summarised in Table 28.

Table 28: Summary of findings by research question

Research question 1: What is the cost of the resources needed to deliver the intervention under the NGO-supported delivery model (constant USD 2023)?					
	Parenting only	Parenting + UCT	UCT only (high)	UCT only (medium)	UCT only (low)
Total programme cost (cost per target child)	829.69	1,075.97	319.63	252.37	146.76
- o/w Investment costs	277.87	279.37	1.50	1.50	1.50
- o/w Recurrent costs	540.44	784.86	317.15	249.90	144.69
- o/w Donated costs	11.37	11.73	0.58	0.58	0.58
Total recurrent provider cost (cost per target child per month)	38.60	56.06	22.65	17.85	10.33
Total recurrent programme cost (cost per target child per month)	39.06	56.54	22.80	18.00	10.38

Note: Provider implementation costs exclude programme design and set-up costs, as well as initial trainings conducted prior to commencement of implementation in Jan 2023

Research question 2: What are the main cost drivers?					
	Parenting only	Parenting + UCT	UCT only (high)	UCT only (medium)	UCT only (low)
Top 3 cost components	Training, 31% Indirect mgt,	Training, 24% Indirect mgt, 22%	Transfers, 77% Indirect mgt, 15%	Transfers, 71% Indirect mgt, 19%	Transfers, 51% Indirect mgt, 33%

	23% Direct Prgr mgt, 21%	Direct prgr mgt, 18%	Direct prgr mgt, 7%	Direct prgr mgt, 9%	Direct prgr mgt, 15%
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Research question 3: How is the cost burden distributed among sources?					
	Parenting only	Parenting + UCT	UCT only (high)	UCT only (medium)	UCT only (low)
NGO	97%	96%	~100%	~100%	100%
Government	3%	4%	-	-	-
Donated (by CHW/CG)	<1%	<1%	<1%	<1%	<1%

Research question 4: What is the incremental cost of the intervention compared to the counterfactual (i.e. the cost of relevant services being delivered in the control group areas)? Incremental recurrent cost per child per month (constant 2023 USD)					
	Parenting only	Parenting + UCT	UCT only (high)	UCT only (medium)	UCT only (low)
Lower bound	37.17	54.70	22.72	17.90	10.38
Upper bound	37.33	54.86			

Some implications of these findings for sustainability, scalability and cost-efficiency of the programme are as follows.

Comparing the treatment arms, the UCT is the least expensive while combined parenting/UCT is most expensive — but these costs need to be considered alongside their relative impacts

On a per child basis, the highest payment UCT arm is nearly two and a half times cheaper to implement than the combined arm – and this does not change much when considering incremental costs. This means that combined parenting/UCT will need to be delivering far better outcomes for children than the cash alone to make this a preferred investment choice, all other things being equal.

The cost per child will need to be brought down substantially in a government-delivery model.

As expected with a small-scale pilot of an NGO-supported model, the cost per child of KK is far higher than estimates of the government budget for ECD-sensitive services per child. Increasing the scale of KK interventions will reduce some costs, but key elements of the KK model will also need to be delivered more cost-efficiently if it is to be sustainable.

The findings shed light on the main cost drivers of each treatment arm, which can help focus efforts on working out cost-efficient and affordable government delivery models.

For example, the two main cost components of the parenting intervention are CHW and HCW training and NGO indirect management costs. While intensive, residential training is an effective modality for developing CHW and HCW skills, alternative approaches may be able to achieve similar results at lower cost. Exploring alternatives, such as more localised, non-residential sessions, along with digital learning tools or blended training approaches, could help reduce costs

while maintaining effectiveness. On the indirect management costs, given the scale of government relative to the NGOs, it seems likely that comparable indirect management services per child would be far cheaper.

The digital app component of the parenting intervention, under the NGO-model, has resource requirements that may be hard to replicate under a government-delivery model.

Maintaining the KK app has required considerable ongoing technical support from NGO specialist staff as well as monthly spending on servers, data bundles, and up-front provision of smart phones (and replacements).

Although CHW stipends and HCWs transport allowances have been estimated to more than cover their time and costs given the dosage of services delivered on average, raising cost-efficiency of this spending is unlikely to be straightforward. One issue is that the stipend and allowances are paid at a flat rate regardless of the type of community being served. In some communities, particularly those where the population is more dispersed, the compensation that CHWs and HCWs receive may not cover their costs. Even over the relatively short implementation period of the KK pilot, retention of CHWs has been challenging, which may imply that they may be in need of additional support. Realising the benefits from the considerable investment in initial training of CHWs relies on retaining them in the programme.

Next steps

The next phase of the costing study involves two key areas of work:

- undertaking a final costing study of the parenting component of KK over its full implementation period (up to end of March 2025)
- estimating the costs of a government delivery model to replicate and scale up the KK parenting and UCT interventions.

These steps are necessary to feed into cost-effectiveness calculations to help understand whether it is worth investing in these interventions, and to propose feasible ways of integrating KK into national systems.

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Appendix A: Field sampling strategy

The KK costing study aims to estimate upper and lower bounds of costs for all four arms of the programme (parenting, UCT, combined parenting/UCT and control). These costs include provider costs (obtained from EGPAF and D-Tree) as well as costs incurred by direct programme stakeholders (CHWs, HCWs and caregivers). The sampling strategy for direct programme stakeholders aimed to identify districts, HDs and households that could provide these different cost estimate bounds. Following a pilot study conducted in Chamwino DC, the following two dimensions were identified as the main drivers of direct stakeholder costs:

- rurality – whether a district is relatively urban or rural affects prices and travel distances
- distance to HD – whether a household is close or far from a HD affects travel costs incurred by CHWs and HCWs as well as, in some cases, by caregivers.

We used the following strategy to identify our target districts, HDs and caregivers.

Districts

We purposively sampled two districts: one urban district (Dodoma County Council) and one rural district (Chemba District Council). These districts were identified as having strongly contrasting rural and urban characteristics and the research team had carried out the KK process evaluation fieldwork in them (for similar reasons) and had built good relationships with district health officials.⁶⁰

Health dispensaries, community health workers and healthcare workers

We sampled four HDs per district – one per programme arm, a total of eight HDs. We used the relative geographical spread of households linked to a HD as a dimension of distance. We used the KK impact evaluation survey baseline dataset of households and HDs that had GPS coordinates to determine geographical spread. These coordinates were plotted on a map and we approximately determined whether a particular HD had a large or

⁶⁰ The KK Impact Evaluation Baseline Study collected a range of relevant contextual variables on each of the eight districts in the programme and these results helped to inform our choice of districts.

small geographical spread. In both districts, for each programme arm, we identified one HD with a relatively large spread of households and one with a relatively small spread of households. From this set of 16 HDs we then selected 8 (split evenly across the districts) such that, across the 4 programme arms, we had one with large geographical spread and one with small geographical spread. This choice was partly informed by the availability of suitable HCWs and CHWs to interview (as explained below). **Table A.1** shows the distribution of sampled HDs.

Table A.1: Distribution of sampled health dispensaries by district, programme arm and geographical spread of households with caregivers

District	Programme arm	Geographical spread of households served by the sampled HD
Dodoma CC	Control	Small
Dodoma CC	Parenting only	Small
Dodoma CC	Parenting + UCT	Large
Dodoma CC	UCT Only	Large
Chemba DC	Control	Large
Chemba DC	Parenting only	Large
Chemba DC	Parenting + UCT	Small
Chemba DC	UCT Only	Small

Source: KK costing study team

Given the small sample size that was possible for this field survey, we needed to speak with CHWs and HCWs who had been in the programme for sufficient time to answer questions around costs incurred (relevant only to parenting arms). We therefore only selected those HDs where the CHW had worked for at least 12 months and the HCW had worked for at least 4 months. As a result, we had to replace two dispensaries that we had initially selected in Chemba District Council since they failed to meet this requirement.

Households with caregivers

We sampled three households with KK caregivers per HD. These households were identified based on their relative distance from the HD – one that was nearest, one that was furthest and another that was mid-way. We sampled three additional households for each facility according to distance, as replacements.

Note on replacements

In total, we replaced two HDs for the reason given above. We also replaced two CHWs because the initial information we had on their status turned out to be incorrect in practice. We also replaced six caregivers, mainly because they were not available to be interviewed. In each case, we followed a prescribed replacement protocol. The KK Costing Study Fieldwork and Data Report (Thrive 2024) has full details.

Appendix B: Top-down costing apportionment assumptions

EGPAF spending on KK

About 40% of EGPAF spending on KK (up until to February 2024) has been allocated between the two interventions (parenting and UCT) and across cost categories using top-down apportionment rates. There are two types of apportionment rates - one based on staff level of effort and the other based on travel activities. Table B.1 shows which apportionment basis has been applied to each of the spending lines. The apportionment rates are in Table B.2 and Table B.3

Table B.1: EGPAF spend on Kizazi Kijacho allocated by top-down apportionment

Spending lines	Apportionment basis		Share of total EGPAF spend (%)
	Staff level of effort	Travel	
Salaries and fringe benefits*	X		18%
Domestic travel		X	4%
Other indirect costs/overheads	X		5%
Indirect costs - fixed percentage	X		13%
Total costs allocated via top-down apportionment			40%

Note: * This spending line also includes the relatively small amount spent on consultants (0.5% of total EGPAF spend).

Source: EGPAF financial statement (as at February 2024)

Table B.2: Apportionment rates based on staff level of effort – by intervention, cost category and year

KK intervention	Cost category	Year 1 – 2021	Year 2 – 2022	Year 3 – 2023	Year 4 – 2024		
		Sep–Dec	Jan–Aug	Sep–Dec	Jan–June	Jul–Dec	Jan–Feb
Assumptions [†]		A	A	B1 B2	C	D	E
Parenting	Prog design	90%	90%				
	Indirect mgt			13%	13%	13%	13%
	Training			60%	7%	10%	34%
	Direct prog mgt				30%	50%	30%
UCT	Prog design	10%	10%				
	Indirect mgt			13%	13%	13%	13%
	Direct prog mgt			14%	37%	14%	10%
All		100%	100%	100%	100%	100%	100%
– Parenting		90%	90%	73%	50%	73%	77%
– UCT		10%	10%	27%	50%	27%	23%

Notes: † Rationale for apportionment: (A) Split 90 (parenting) to 10 (UCT) because parenting is a much more complicated intervention to design. (B1) In the original EGPAF budget the local staff costs were split 74% to programme staff and 26% to operations staff. Same split applied here such that indirect management is allocated 26%. (B2) Parenting training with two full-time programme officers and ECD coordinator heavily involved. On the UCT, beneficiaries were phoned, a 1-day staff orientation and registration started in December. The 74% for programme staff was split 60% for parenting training and 14% for UCT. (C) Parenting home visits started, the programme officers started supervision, and refresher training in April. On the UCT, registration continued until early June and first payments started in March, temporary staff were taken on for a couple of months. Hence the 74% was split evenly between parenting (further split into 7% for training and 30% for supervision/management based on approximate time for training) and UCT. (D) Parenting home visits continued, group session training, and group sessions started. On the UCT, payments continued. The 74% was split 60% to parenting (10% training and 50% supervision/management) and 14% to UCT. (E) Parenting home visits/group sessions continued and second training took place. On the UCT, the final payments were made.

Table B.3: Apportionment rates based on travel activities – by intervention, cost category and year

KK intervention	Cost category	Year 1 –2021	Year 2 - 2022	Year 3 - 2023	Year 4 - 2024		
		Sep– Dec	Jan– Aug	Sep– Dec	Jan– June	Jul– Dec	Jan– Feb
Assumptions [†]		A	A	B	C	D	E
Parenting	Prog design	90%	90%				
	Training			95%	10%	15%	50%
	Direct prog mgt				60%	80%	45%
UCT	Prog design	10%	10%				
	Direct prog mgt			5%	30%	5%	5%
All		100%	100%	100%	100%	100%	100%
– Parenting		90%	90%	95%	70%	95%	95%
– UCT		10%	10%	5%	30%	5%	5%

Notes: † Rationale for apportionment: (A) Based on assumed split of staff LOE (see Table B.2).

(B) Assumed split of staff level of effort between parenting/training and UCT/direct programme management is ~80%:20% (see Table B.2), EGPAF advised split of travel was more skewed to parenting/training at 95%: 5%.

(C) Assumed split of staff level of effort between parenting/training, parenting/direct programme management and UCT/direct programme management is ~10%:40%:50% (see Table B.2). EGPAF advised split of travel more skewed at 10%:60%:30%

(D) Assumed split of staff level of effort between parenting/training, parenting/direct programme management and UCT/direct programme management is ~15%:70%:15% (see Table B.2). EGPAF advised more realistic split of 15%:80%:5%

(E) Assumed split of staff level of effort between parenting/training, parenting/direct programme management and UCT/direct programme management is ~ 50%, 40%, 10% (see Table B.2). EGPAF advised a more realistic split is 50%:45%:5%.

D-tree spending on KK

As Table B.4 shows, a relatively high share of D-tree spending (85%) has been allocated using a top-down apportionment method. D-tree was responsible for the digital app, and much of its spending is on labour (57%) needed for a range of IT development and management activities that does not lend itself to bottom-up costing. The other major component of spending is the fixed percentage (15%) charged for indirect costs.

Table B.4: D-tree spend on Kizazi Kijacho allocated by top-down apportionment

Spending lines	Share of total D-tree spend
Labour	57%
Materials	2%
Travel	7%
Facilities	3%
Indirect costs -fixed %	15%
Total spend allocated top-down	85%

Source: D-tree financial statement (as at February 2024)

D-tree worked on the parenting component of KK only, not the UCT, so there is no need to apportion spending between the two interventions. Table B.5 shows how D-tree spending on the parenting intervention has been assigned to the relevant cost categories and years.

Table B.5: Mapping of D-tree spending lines to cost category, resource type and year

D-Tree Budget Category		PROG. DESIGN		INDIRECT PROG. M/MENT			TRAINING			DIRECT PROG. M/MENT		
		2021	2022	2022	2023	2024	2022	2023	2024	2022	2023	2024
line item	Resource type	Mar-Dec	Jan-Aug	Sep-Dec	Jan-Dec	Jan-Feb	Sep-Dec	Jan-Dec	Jan-Feb	Sep-Dec	Jan-Dec	Jan-Feb
Finance, Operations, & Administration												
Staff salaries & fringe benefits	Labour	✓	✓	✓	✓	✓						
Staff equipment & supplies	Materials		✓	✓	✓	✓						
Rent and utility costs	Facilities	✓	✓	✓	✓	✓						
Pro-rated operating costs	Materials	✓	✓	✓	✓	✓						
Project & System Design and Requirements gathering												
Staff salaries & fringe benefits	Labour	✓										
System Set-up and Development (Infrastructure)												
Technology vendor (Ona)	Labour	✓	✓							✓		
System Development & Testing												
Staff salaries & fringe benefits	Labour	✓	✓									
Travel & Transportation	Travel	✓	✓									
Training and Pilot												
Staff salaries & fringe benefits	Labour						✓					
Travel & Transportation	Travel						✓					
Implementation, Performance Monitoring & Evaluation, and Maintenance												
Staff salaries & fringe benefits	Labour							✓	✓	✓	✓	✓
Travel & Transportation	Travel							✓	✓	✓	✓	✓
Meetings and Trainings	Travel										✓	✓
System maintenance (Infrastructure)												
Technology vendor (Ona)	Labour									✓	✓	✓
Indirect costs												
Fixed % of direct costs	Other	✓	✓	✓	✓	✓						

Notes: 1. Staff salaries and fringe benefits include the cost of consultants (e.g. IT engineering time, which was outsourced). 2. Under the D-tree budget category 'implementation, performance monitoring and evaluation and maintenance': (i) staff remuneration costs were split between training and direct management based on the share of training days in a particular month (ii) travel costs were allocated entirely to the training category during the months when training took place.

Appendix C: Further cost details

Table C.1: Total programme cost of UCT only – low payment intervention by cost category (constant 2023 USD)

Cost category		Investment	Recurrent	Both	Share of total provider cost (%)
		Total	Total	Total	
Total provider costs		226.92	21,863.76	22,090.68	100%
Overhead costs		226.92	7,297.83	7,524.75	34%
	Indirect management	-	7,297.83	7,297.83	33%
	Prog design	226.92	-	226.92	1%
Direct costs		-	14,565.93	14,565.93	66%
	Training	-	-	-	0%
	Direct delivery	-	-	-	0%
	Direct prog mgt	-	3,345.92	3,345.92	15%
	Transfers	-	11,220.01	11,220.01	51%
Total provider cost per child		1.50	144.79	146.30	
Total provider cost per month per child		0.11	10.34	10.45	
Total donated costs		-	87.21	87.21	0.4%
	Caregiver time (during UCT registration)	-	87.21	87.21	
Total donated cost per child				0.58	
Total donated cost per month per child			0.04		
Total overall programme costs		226.92	21,950.97	22,177.897	
	Total cost per child	1.50	145.37	146.87	
	Total cost per month per child	0.11	10.38	10.49	

Table C.2: Total programme cost of UCT only – medium payment intervention by cost category (constant 2023 USD)

Cost category	Investment	Recurrent	Both	Share of total provider cost (%)
	Total	Total	Total	
Total provider costs	249.46	41,500.67	41,750.13	100%
Overhead costs	249.46	8,022.78	8,272.24	20%
Indirect management	-	8,022.78	8,022.78	19%
Prog design	249.46	-	249.46	1%
Direct costs	-	33,477.89	33,477.89	80%
Training	-	-	-	0%
Direct delivery	-	-	-	0%
Direct prog mgt	-	3,678.29	3,678.29	9%
Transfers	-	29,799.60	29,799.60	71%
Total provider cost per child	1.50	250.00	251.51	
Total provider cost <u>per month</u> per child	0.11	17.86	17.96	
Total donated costs	-	95.88	95.88	0.2%
Caregiver time (during UCT registration)	-	95.88	95.88	
Total donated cost per child			0.58	
Total donated cost <u>per month</u> per child		0.04	0.04	
Total overall programme costs	249.46	41,596.55	41,846.01	
Total cost per child	1.50	250.08	252.08	
Total cost <u>per month</u> per child	0.11	17.90	18.01	

Table C.3: Total programme cost of UCT only – high payment intervention by cost category (constant 2023 USD)

Cost category	Investment	Recurrent	Both	Share of total provider cost (%)
	Total	Total	Total	
Total provider costs	1,384.04	292,439.65	293,823.70	100%
Overhead costs	1,384.04	44,511.93	45,895.97	16%
Indirect management	-	44,511.93	44,511.93	15%
Prog design	1,384.04	-	1,384.04	0%
Direct costs	-	247,927.72	247,927.72	84%
Training	-	-	-	0%
Direct delivery	-	-	-	0%
Direct prog mgt	-	20,407.87	20,407.87	7%
Transfers	-	227,519.85	227,519.85	77%
Total provider cost per child	1.50	317.52	319.03	
Total provider cost <u>per month</u> per child	0.11	22.68	22.79	
Total donated costs	-	532.10	532.10	0.2%
Caregiver time (during UCT registration)	-	531.95	532.10	
Total donated cost per child			0.58	
Total donated cost <u>per month</u> per child		0.04	0.04	
Total overall programme costs	1,384.04	281,054.91	294,719.67	
Total cost per child	1.50	318.10	319.60	
Total cost per month per child	0.11	22.72	22.83	

Table C.4: Total provider cost of parenting interventions (all) by cost category and resource type by year (constant 2023 USD)

Cost category and resource type		Investment	Recurrent	Both	Share of total (%)	Total cost per child
		Total	Total	Total		
Programme design		214,244.21	-	214,244.21	100%	144.08
	Facilities	5,910.19	-	5,910.19	3%	
	Labour	157,825.75	-	157,825.75	74%	
	Materials and equipment	13,354.79	-	13,354.79	6%	
	Travel and accommodation	4,742.57	-	4,742.57	2%	
	Other	32,410.91	-	32,410.91	15%	
Indirect management		-	277,735.38	277,735.38	100%	186.78
	Facilities	-	23,366.82	23,366.82	8%	
	Labour	-	57,552.67	57,552.67	21%	
	Materials and equipment	-	18,049.85	18,049.85	6%	
	Travel and accommodation	-	1,852.68	1,852.68	1%	
	Other	-	176,913.35	176,913.35	64%	
Training		167,270.39	208,309.42	375,579.81	100%	252.58
	Facilities	2,612.23	1,329.90	3,942.13	1%	
	Labour	23,165.28	83,870.07	107,035.35	28%	
	Materials and equipment	-	1,239.10	1,239.10	0%	
	Travel and accommodation	141,492.88	121,870.35	263,363.23	70%	
	Other	-	-	-	0%	
Direct delivery		25,969.87	71,439.89	97,409.75	100%	65.51
	Facilities	-	-	-	0%	
	Labour	-	47,293.06	47,293.06	49%	
	Materials and equipment	25,969.87	9,958.91	35,928.78	37%	
	Travel and accommodation	-	14,187.92	14,187.92	15%	
	Other	-	-	-	0%	
Direct programme management		6,826.22	248,317.39	255,143.61	100%	171.58
	Facilities	-	-	-	0%	
	Labour	-	170,434.96	170,434.96	67%	
	Materials and equipment	6,826.22	24,246.15	31,072.37	12%	
	Travel and accommodation	-	53,636.28	53,636.28	21%	
	Other	-	-	-	0%	

Table C.5: Comparison of top-down and bottom-up estimates of community health worker and healthcare worker delivery costs (constant 2023 USD)

			Investment	Recurrent	Both
Total CHW stipends paid			-	47,293.06	47,293.06
Total CHW stipend cost (bottom-up)			-	26,938.87	26,938.87
	Labour	Home visits	-	13,762.87	13,762.87
	Transport	Home visits	-	11,674.10	11,674.10
	Labour	Group sessions	-	1,501.90	1,501.90
	Transport	Group sessions	-	-	-
<i>Bottom-up as % of actual paid</i>			-	57%	57%
Total HCW travel allowances paid			-	14,187.92	14,187.92
Total HCW travel allowance (bottom-up)			-	1,618.87	1,618.87
	Transport	Home visits	-	1,027.91	1,027.91
	Transport	Group sessions	-	590.96	590.96
<i>Bottom-up as % of actual paid</i>			-	11%	11%

Notes: (i) Actual dosage of home visits, group sessions and caregiver attendance at group sessions is from KK app data (based on 154 CHS, 88 HCWs, 1,350 registered caregivers without app problems). (ii) Average duration of home visits and of group sessions are from the Costing Field Survey Data (median of responses from caregivers and CHWs). (iii) Control group estimate of home visits is partly from the KK Phase 1 Impact Evaluation data (antenatal and postnatal care home visits). The survey did not cover child home visits (four would have taken place if the government protocol had been followed) and so we estimated an upper bound and lower bound for child home visits and added this to the estimated antenatal and postnatal home visits. Details are in Table 18.

Sources: KK programme design documents; KK app data; costing field survey; KK Phase 1 impact evaluation survey.

Box C.1: Control area cost estimation

A. Average intensity ratio of home visits in control compared to treatment (from Table 18)

Lower bound = $0.6/10.6 = 5.66\%$

Upper bound = $1.4/10.6 = 13.21\%$

B. Total CHW stipends paid for home visits in treatment areas

CHW stipends are meant to be for both home visits and group sessions and hence, the financial statements only show total stipends paid out. However, through our bottom-up costing methodology, we have separate estimates of CHW costs for home visits and for group sessions. Thus, we apportion the total CHW stipends by this ratio of home visits to group estimates obtained from the bottom-up estimates (refer to Table C.4).

CHW total stipends paid (top-down estimate) = USD 47,293.06

Ratio of CHW home visit to CHW total (bottom-up) =
 $((25,436.97 + 1,501.90) / (26,938.87)) = 94\%$

Thus, CHW total stipends paid for home visits = USD 44,656.37

C. Control area cost estimates (A x B) = USD 2,527.72 to USD 5,898.01

D. Per child control cost estimates

Number of control caregivers (targeted) = 856

Per child total control costs = USD 2.95 — USD 6.89 per child (USD 0.21 — USD 0.49 per child per month)

Appendix D: Data sources for unit costs and dosage

Table D.1: Data sources for unit costs

Resource type	Unit	Source	Notes
1. Parenting			
Labour			
CHW stipend	USD/month-person	KK Programme Design Document	Set rate of TZS 50,000 per month, fixed in nominal terms over implementation period.

	CHW volunteer time	USD/hour	KK Impact Evaluation Baseline Survey (2022)	Median hourly wage (TZS 1,200) reported by CHWs for non-CHW work (in 2022 prior to KK implementation). Note that the median TZS 1,200 was much lower than the mean TZS 4,379. The rate was adjusted for inflation where the cost occurred in a different year to the initial price.
	HCW salary	USD/month-person	Government salary scale	TGHS-B scales (middle of enrolled nurse and nursing officer) = TZS 740,000pm + 15% employer contributions. Converted to a monthly rate by dividing by 12. We have made no inflation adjustments.
	Regional/ District govt official salary	USD/month-person	Government salary scale	TGHS-C scale (CCHWCO with degree) = TZS 1,040,000pm + 15% employer contribution. Converted to a monthly rate. We have made no inflation adjustments.
	CG time	USD/hour	National Bureau of Statistics. Tanzania Integrated Labour Force Survey 2020/21	Estimate of opportunity cost of CG time using the Female median agricultural wage of TZS 64,000 per month (2021 prices). Adjusted for inflation where the cost occurred in a different year to the initial price.
Materials and equipment				
	Smart phones for CHWs/HCWs	USD/phone	D-Tree finance team	Purchased prior to implementation with a contingency number for replacements. Weighted average of actual purchase prices where different batches had slightly different costs.
	Solar chargers for CHWs	USD/charger	D-Tree finance team	Purchased prior to implementation with a contingency number for replacements. Weighted average of actual purchase prices where different batches had slightly different costs.
	Comms bundle for CHWs/HCWs	USD/month-person	D-Tree finance team	Weighted average of the amount paid per bundle to two different providers based on numbers of CHWs/HCWs using the providers.
	Material kit for CHWs	USD/kit	EGPAF KK expenditure statement	Based on total spending on materials that were given to CHWs divided by the number of CHWs.
	Server hosting	USD/month	D-Tree finance team	Sum of monthly server hosting costs for three different platforms (Ona, AWS and Sisense).
Travel and accommodation				
	Training – DSA	USD/night-person	Interview with EGPAF Senior ECD manager	The lower rate is for CHWs and the higher rate for HCWs and other officials. The rates have not changed throughout the programme.
	Training – Conference package	USD/day-person	EGPAF KK budget statement	
	Training – transport	USD/return journey	EGPAF KK budget statement	
	Supervision – transport	USD/month - CHW	KK Programme Design Document	Set rate of TZS 15,000 per month for each CHW that the HCW supervises (1.8 CHW per HCW on average), fixed in

	allowance for HCWs			nominal terms over the implementation period.
	Quarterly supervision – DSA for govt officials	USD/night-person	Government DSA rates	For TGHS-B and TGHS-C scale the overnight DSA rate is TZS 130,000 to cover accommodation and food. The rate has stayed the same in nominal terms since 2022.
Facilities				
	Training – venue hire	USD/day	EGPAF KK budget statement	
	Group session – room rental	USD/hour	Study team	Holding estimate of TZS 1000 per hour (~USD 3 per day)
2. UCT				
	Cash transfer – low amount	USD/2 months-person	KK Programme Design Document	Set rate of TZS 32,000 plus TZS 200 bank transfer fee per month, fixed in nominal terms over implementation period.
	Cash transfer – medium amount	USD/2 months-person	KK Programme Design Document	Set rate of TZS 77,000 plus TZS 200 bank transfer fee per month, fixed in nominal terms over implementation period.
	Cash transfer – high amount	USD/2 months-person	KK Programme Design Document	Set rate of TZS 109,000 plus TZS 200 bank transfer fee per month, fixed in nominal terms over implementation period.

Appendix E: Additional hypothetical scaling scenario costs

Table E.1 below represents the provider costs per child per month when the programme is scaled as per current dosage using the NGO-supported model. It also includes provider costs when scaling a hypothetical intervention of Parenting and UCT where the UCT amount is high.

Table E.1: Provider costs when scaling parenting intervention arms at current dosage

# of CGs	CGs per CHW	# of cash transfers	Scenario details	Provider costs (per child, per month)		
				Investment	Recurrent	Total
1353	8.79	0	Current scenario	19.90	38.71	58.61
1487	9.66	0	As per KK design	19.90	38.71	58.61
2427	15.76	0	Max CHW stipend used	12.19	23.71	35.91
4552	29.56	0	Max CHW time-used	6.50	12.64	19.14
7885	51.20	0	Provider Recurrent cost of Parenting and UCT is same as that of UCT only – high	3.75	7.30	11.05
726	9.43	642	Current scenario	20.01	56.21	76.22

1487	9.66	1487	As per KK design	20.04	59.45	79.50
2427	15.76	2427	Max CHW stipend used	12.28	41.89	54.17
4552	29.56	4552	Max CHW time-used	6.55	28.91	35.46
7885	51.20	7885	Provider recurrent cost is USD 22.65 (same as that of UCT only – high)	3.78	22.65	26.43
1353	8.79	1353	Hypothetical current	22.03	69.74	91.77
1487	9.66	1487	As per KK design	20.04	65.25	85.29
2427	15.76	2427	Max CHW stipend used	12.28	47.68	59.96
4552	29.56	4552	Max CHW time-used	6.55	34.71	41.25

Source: Authors' calculations

Thrive

Thrive is a multi-country research programme that aims to support countries to turn what we know about positive early childhood development into practical, scalable, low-cost programmes, able to transform societies over multiple generations. Working closely with policymakers and other stakeholders, Thrive aims to build understanding of early childhood development service delivery models and how they can be provided cost effectively and at scale, and how these systems can innovate, improve, and better serve children and communities in low- and middle-income countries.

Our five focus countries are Bangladesh, Ghana, Kiribati, Sierra Leone and Tanzania.

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